Week 2. <u>Basics about model theory</u>

<u>Def 1.1</u>: A <u>language</u> $\mathcal{L}$ is a set of constants, function symbols and relation symbols.

<u>Rem 1.2</u>: For each function symbol (or relation symbol), there is an arity associated to them. The arity of each function symbol f (resp. relation symbol R) is an integer $n \in \mathbb{N}$ s.t. f ~~has~~ has n variables (resp. R is an n-ary relation).

<u>Example 1.3</u>: (1) $\mathcal{L}_\phi = \phi$, the empty language.

(2) $\mathcal{L}_{AbG} = \{0, +, -\}$, the language of abelian groups.

(3) $\mathcal{L}_{Ring} = \{0, +, -, 1, \cdot\}$, the language of rings with unit.

(4) $\mathcal{L}_{Group} = \{e, \cdot, ^{-1}\}$, the language of groups.

(5) $\mathcal{L}_{order} = \{<\}$, the language of orders.

(6) $\mathcal{L}_{ORing} = \mathcal{L}_{Ring} \cup \mathcal{L}_{order}$

Constants: 0, 1, e
Unitary function symbols: $^{-1}$
Binary function symbols: $+, -, \cdot$
Binary relation symbol: $<$

<u>Def 1.4</u>: Let $\mathcal{L}$ be a language. An <u>$\mathcal{L}$-structure</u> $\mathcal{M}$ is given by the following data:

i). a nonempty set M called the <u>universe or domain</u>.

ii). a function $f^{\mathcal{M}}: M^{n_f} \to M$ for each function symbol $f \in \mathcal{L}$ with arity n_f.

iii). a set $R^{\mathcal{M}} \subseteq M^{n_R}$ for each relation symbol $R \in \mathcal{L}$ with arity n_R.

iv). an element $c^{\mathcal{M}} \in M$ for each constant symbol $c \in C$.

The cardinality of a structure is the cardinality of its universe.

Def 1.5: Let $\mathcal{M}$ and $\mathcal{N}$ be $\mathcal{L}$-structures with universe M and N respectively. A map $h: M \to N$ is called a homomorphism if for all $x_1, \cdots, x_n \in M$

$$h(c^{\mathcal{M}}) = c^{\mathcal{N}}$$
$$h(f^{\mathcal{M}}(x_1, \cdots, x_n)) = f^{\mathcal{N}}(h(x_1), \cdots, h(x_n))$$
$$R^{\mathcal{M}}(x_1, \cdots, x_n) \Rightarrow R^{\mathcal{N}}(h(x_1), \cdots, h(x_n)).$$

for all constants c, n-ary function symbols f and relation symbols R from $\mathcal{L}$. We denote this by $h: \mathcal{M} \to \mathcal{N}$.
If in addition h is injective and $R^{\mathcal{M}}(x_1, \cdots, x_n) \Leftrightarrow R^{\mathcal{N}}(h(x_1), \cdots, h(x_n))$ for all $x_1, \cdots, x_n \in M$, then h is called an embedding. An isomorphism is a surjective embedding.

Rem 1.6: The set of automorphisms of an $\mathcal{L}$-structure forms a group under composition.

Def 1.7: We call $\mathcal{M}$ a substructure of $\mathcal{N}$ if $M \subseteq N$ and if the inclusion map is an embedding from $\mathcal{M}$ to $\mathcal{N}$. We denote this by $\mathcal{M} \subseteq \mathcal{N}$. We say $\mathcal{N}$ is an extension of $\mathcal{M}$ if $\mathcal{M}$ is a substructure of $\mathcal{N}$.

Rem 1.8: If $\mathcal{N}$ is an $\mathcal{L}$-structure and M a non-empty subset of N, then M is the universe of a (uniquely determined) substructure $\mathcal{M}$ iff. M contains all $c^{\mathcal{N}}$ and M is closed under all operations $f^{\mathcal{N}}$. In particular, If $\mathcal{L}$ does not contain any constants or function symbols, then any non-empty subset of an $\mathcal{L}$-structure is again an $\mathcal{L}$-structure. Also, if $h: \mathcal{M} \to \mathcal{N}$, then $h(M)$ is the universe of a substructure of $\mathcal{N}$.

Def 1.9: An <u>$\mathcal{L}$-term</u> is a word (sequence of symbols) built from constants, the function symbols of $\mathcal{L}$ and the <u>variables</u> $v_0, v_1, \cdots$ according to the following rules:

(1) Every variable v_i and every constant c is an $\mathcal{L}$-term.

(2). If f is an n-ary function symbol and $t_1, \cdots, t_n$ are $\mathcal{L}$-terms, then $f(t_1, \cdots, t_n)$ is also an $\mathcal{L}$-term.

The number of occurences of function symbols in a term is called its <u>complexity</u>.

Example 1.10: $\cdot + xy + zw = (x+y)\cdot(z+w)$

Def 1.11: Let $\mathcal{M}$ be an $\mathcal{L}$-structure. A sequence of elements $\vec{b} = (b_0, b_1, \cdots)$ could be considered as <u>assignments</u> to the variables $v_0, v_1, \cdots$. We define the <u>interpretation</u> $t^{\mathcal{M}}[\vec{b}]$ for any $\mathcal{L}$-term t recursively as:

$$v_i^{\mathcal{M}}[\vec{b}] = b_i, \quad c^{\mathcal{M}}[\vec{b}] = c^{\mathcal{M}}, \quad f(t_1, \cdots, t_n)[\vec{b}] = f^{\mathcal{M}}(t_1^{\mathcal{M}}[\vec{b}], \cdots, t_n^{\mathcal{M}}[\vec{b}])$$

Some symbols:

Auxiliary symbols: "(", and ")".

Logical symbols: Variables $v_0, v_1, \cdots$
equality symbol $\doteq$
negation symbol $\neg$
conjunction symbol $\wedge$
existential quantifier $\exists$

Def 1.12: $\mathcal{L}$-formulas are:

(1) $t_1 \doteq t_2$ where t_1, t_2 are $\mathcal{L}$-terms.

(2) $R(t_1, \cdots, t_n)$ where R is an n-ary relation symbol from $\mathcal{L}$ and $t_1, \cdots, t_n$ are $\mathcal{L}$-terms

(3). $\neg\psi$ where ψ is an $\mathcal{L}$-formula.

(4) $(\psi_1 \wedge \psi_2)$ where ψ_1 and ψ_2 are $\mathcal{L}$-formulas

(5) $\exists x\, \psi$ where ψ is an $\mathcal{L}$-formula and x a variable.

Formulas of the form $t_1 \doteq t_2$ or $R(t_1, \cdots, t_n)$ are called atomic

The complexity of a formula is the number of occurrences of $\neg, \exists$ and $\wedge$.

Some abbreviations:

$$(\psi_1 \vee \psi_2) = \neg(\neg\psi_1 \wedge \neg\psi_2)$$
$$(\psi_1 \to \psi_2) = \neg(\psi_1 \wedge \neg\psi_2)$$
$$(\psi_1 \leftrightarrow \psi_2) = \big((\psi_1 \to \psi_2) \wedge (\psi_2 \to \psi_1)\big)$$
$$\forall x\, \psi = \neg \exists x \neg \psi.$$

Def 1.13: Let $\mathcal{M}$ be an $\mathcal{L}$-structure. For $\mathcal{L}$-formula φ and assignments $\vec{b}$ we define the relation $\mathcal{M} \models \varphi[\vec{b}]$ recursively over φ:

$\mathcal{M} \models t_1 \doteq t_2[\vec{b}] \iff t_1^{\mathcal{M}}[\vec{b}] = t_2^{\mathcal{M}}[\vec{b}]$

$\mathcal{M} \models R(t_1, \cdots, t_n)[\vec{b}] \iff R^{\mathcal{M}}(t_1^{\mathcal{M}}[\vec{b}], \cdots, t_n^{\mathcal{M}}[\vec{b}])$.

$\mathcal{M} \models \neg\psi[\vec{b}] \iff \mathcal{M} \not\models \psi[\vec{b}]$

$\mathcal{M} \models (\psi_1 \wedge \psi_2)[\vec{b}] \iff \mathcal{M} \models \psi_1[\vec{b}]$ and $\mathcal{M} \models \psi_2[\vec{b}]$

$\mathcal{M} \models \exists x \psi[\vec{b}] \iff$ there exists $a \in M$ s.t. $\mathcal{M} \models \psi[\vec{b}\frac{a}{x}]$

Here we use the notation $\vec{b}\frac{a}{x} = (b_0, \cdots, b_{i-1}, a, b_{i+1}, \cdots)$ if $x = v_i$.

If $\mathcal{M} \models \varphi[\vec{b}]$ holds we say φ holds in $\mathcal{M}$ for $\vec{b}$ or $\vec{b}$ satisfies φ (in $\mathcal{M}$).

Note: every formula has unique decomposition into subformulas.

Def 1.14: The variable x occurs freely in the formula φ if it occurs at a place which is not within the scope of a quantifier $\exists$. Otherwise its occurrence is called bound. Formal definition:

x free in $t_1 \doteq t_2 \iff x$ occurs in t_1 or in t_2.

x free in $R(t_1, \cdots, t_n) \iff x$ occurs in one of the t_i.

x free in $\neg\psi \iff x$ free in ψ

x free in $(\psi_1 \wedge \psi_2) \iff x$ free in ψ_1 or x free in ψ_2.

x free in $\exists y \psi \iff x \neq y$ and x free in ψ.

Example 1.15: $\forall v_0 (\exists v_1 R(v_0, v_1) \wedge P(v_2))$

v_0, v_1 are bounded.

v_2 free.

Lem 1.16: Suppose $\vec{b}$ and $\vec{c}$ agree on all variables which are free in φ.

Then $\mathcal{M} \models \varphi[\vec{b}] \iff \mathcal{M} \models \varphi[\vec{c}]$.

Pf: Induction on the complexity of φ.

Def 1.17: A formula φ without free variables is called a sentence.

An $\mathcal{L}$-theory T is a set of $\mathcal{L}$-sentences. We write $\mathcal{M} \models \varphi$ if $\mathcal{M} \models \varphi[\vec{b}]$ for some $\vec{b}$.

$\varphi(x_1, \cdots, x_n)$, an $\mathcal{L}$-formula with free variables among $\{x_1, \cdots, x_n\}$.

If $a_1, \cdots, a_n \in M$, we define $\mathcal{M} \models \varphi[a_1, \cdots, a_n]$ by $\mathcal{M} \models \varphi[\vec{b}]$, where $\vec{b}$ is an assignment satisfying $\vec{b}(x_i) = a_i$. Well-defined by Lem 1.16.

Thus, $\varphi(x_1, \cdots, x_n)$ defines an n-ary relation

$$\varphi[\mathcal{M}] = \{\bar{a} \mid \mathcal{M} \models \varphi[\bar{a}]\}.$$

on M, the realisation set of φ. We call two formulas equivalent if in every structure they define the same set.

A formula is in negation normal forms if it is buit from basic formulas using $\wedge, \vee, \exists, \forall$.

Lem 1.18: Every formula can be transformed to an equivalent formula which is in negation normal form.

Pf: $\neg(\varphi\wedge\psi) \sim (\neg\varphi \vee \neg\psi)$, $\neg(\varphi\vee\psi)\sim(\neg\varphi\wedge\neg\psi)$,

$\neg\exists x\varphi \sim \forall x\neg\varphi$, $\neg\forall x\varphi \sim \exists x\neg\varphi$, $\neg\neg\varphi\sim\varphi$. □

Def 1.19: A formula in negation normal form which does not contain any existential quantifier is called universal. Formulas in negation normal form without universal quantifiers are called existential.

Lem 1.20: Let $h: \mathcal{M}\longrightarrow\mathcal{N}$ be a homomorphism ~~iff~~ between $\mathcal{L}$-structures.

(1) If h is an isomorphism, then $\mathcal{M}\models\varphi[a_1,\cdots,a_n] \iff \mathcal{N}\models\varphi[h(a_1),\dots,h(a_n)]$.

(2) Assume h is an embedding. For all existential formulas $\varphi(x_1,\cdots,x_n)$ and all $a_1,\cdots,a_n\in A$ we have

$$\mathcal{M}\models\varphi[a_1,\cdots,a_n] \Rightarrow \mathcal{N}\models\varphi[h(a_1),\cdots,h(a_n)]$$

For universal φ, the dual holds:

$$\mathcal{N}\models\varphi[h(a_1),\dots,h(a_n)] \Rightarrow \mathcal{M}\models\varphi[a_1,\dots,a_n]$$

Pf: Kent & Ziegler, Lemma 1.2.16 □

Def 1.21: Let T be an $\mathcal{L}$-theory. An $\mathcal{L}$-structure $\mathcal{M}$ is a model of T if $\mathcal{M}\models\varphi$ for all $\varphi\in T$. A theory which has a model is a consistent theory. Let $\Sigma :=$ set of $\mathcal{L}$-formulas. We say Σ is consistent if $\mathcal{M}\models\varphi[\vec{b}]$ for all $\varphi\in\Sigma$. We say Σ is consistent with T if $T\cup\Sigma$ is consistent.

$Th(\mathcal{M}) :=$ set of $\mathcal{L}$-sentences φ s.t. $\mathcal{M}\models\varphi$, the full theory of $\mathcal{M}$.

Lem 1.22: Let T be an $\mathcal{L}$-theory and $\mathcal{L}'$ be an extension of $\mathcal{L}$.

Then T is consistent as an $\mathcal{L}$ theory iff. T is consistent as an $\mathcal{L}'$-theory.

Pf: Every $\mathcal{L}$-structure is ~~extend~~ expandable to an $\mathcal{L}'$-structure. □

Example 1.23: (1). Infinite sets: $\mathcal{L} = \phi$. Consider the $\mathcal{L}$-theory where we have, for each n, the sentence φ_n given by

$$\exists x_1 \exists x_2 \cdots \exists x_n \bigwedge_{i<j\leq n} x_i \neq x_j .$$

(2). Linear orders: $\mathcal{L} = \{<\}$, where $<$ is a binary relation symbol. The theory of ~~~~ linear orders is given by sentences:

$$\forall x \neg (x<x), \ \forall x \forall y \forall z ((x<y \wedge y<z) \rightarrow x<z),$$
$$\forall x \forall y (x<y \vee x=y \vee y<x).$$

(3) ~~~~ Abelian Groups. $\mathcal{L} = \{0, +, -\}$. Theory of abelian groups:

$$\forall x \forall y \forall z ((x+y)+z = x+(y+z))$$
$$\forall x (0+x = x)$$
$$\forall x ((-x)+x = 0)$$
$$\forall x \forall y (x+y = y+x).$$

Def 1.24: If a sentence φ holds in all models of T, we say that φ follows from T (or that T proves φ) and write $T \vdash \varphi$.

By lemma 1.22 this relation is independent of the language.

Lem 1.25: (1) If $T \vdash \varphi$ and $T \vdash (\varphi \rightarrow \psi)$, then $T \vdash \psi$.

(2). If $T \vdash \varphi(c_1, \cdots, c_n)$ and the constants $c_1, \cdots, c_n$ occur neither in T nor in $\varphi(x_1, \ldots, x_n)$, then $T \vdash \forall x_1 \cdots \forall x_n \varphi(x_1, \cdots, x_n)$

Pf: Let $\mathcal{L}' = \mathcal{L} \setminus \{c_1, \cdots, c_n\}$. If the $\mathcal{L}'$-structure $\mathcal{M}$ is a model of T and $a_1, \cdots, a_n$ are arbitrary elements, then $(\mathcal{M}, a_1, \cdots, a_n) \models \varphi(c_1, \cdots, c_n)$. That means $\mathcal{M} \models \forall x_1 \cdots \forall x_n \varphi(x_1, \cdots, x_n)$.

Thus $T \vdash \forall x_1 \cdots \forall x_n \varphi(x_1, \cdots, x_n)$ □.

Def 1.26: A consistent $\mathcal{L}$-theory T is called complete if ~~fat~~ for all $\mathcal{L}$-sentences φ, $T \vdash \varphi$ or $T \vdash \neg\varphi$.

This notion depends on $\mathcal{L}$.

$|T| := \max\{|\mathcal{L}|, \aleph_0\}$, it is exactly the number of $\mathcal{L}$-formulas.

$Th(\mathcal{M})$ is a complete theory.

Def 1.27: Two $\mathcal{L}$-structures $\mathcal{M}$ and $\mathcal{N}$ are called elementarily equivalent, $\mathcal{M} \equiv \mathcal{N}$, if they have the same full theory, i.e. $\forall$ $\mathcal{L}$-sentence φ
$$\mathcal{M} \models \varphi \Leftrightarrow \mathcal{N} \models \varphi.$$

Isomorphic structures are always elementarily equivalent.

Lem 1.28: Let T be a consistent theory. Then TFAE:

(1) T is complete.

(2) All models of T are elementarily equivalent.

(3). There exists a structure $\mathcal{M}$ with $T \equiv Th(\mathcal{M})$.

(S: another theory. $T \vdash S$ iff $T \vdash \varphi$ for all $\varphi \in S$. $T \equiv S$ iff $T \vdash S$ and $S \vdash T$)

Pf: (1) $\Rightarrow$ (3): Let $\mathcal{M}$ be a model of T. If φ holds in $\mathcal{M}$, then $T \nvdash \neg\varphi$ and thus $T \vdash \varphi$. So $T \equiv Th(\mathcal{M})$ holds.

(3) $\Rightarrow$ (2): If $\mathcal{N} \models T$, then $\mathcal{N} \models Th(\mathcal{M})$ and therefore $\mathcal{N} \equiv \mathcal{M}$. Note that $\equiv$ is an equivalence relation.

(2) $\Rightarrow$ (1): Let $\mathcal{M}$ be a model of T. If φ holds in $\mathcal{M}$, then φ holds in all models of T, i.e. $T \vdash \varphi$. Otherwise, $\neg\varphi$ holds in $\mathcal{M}$ and we have $T \vdash \neg\varphi$ □

Model Theory

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2 Elementary Extensions and The Compactness Theorem

Definition 2.1. Let $\mathcal{N}$ and $\mathcal{M}$ be two L-structures. A map $h : \mathcal{N} \to \mathcal{M}$ is called elementary if it preserves the validity of arbitrary formulas $\varphi(x_1, \dots, x_n)$. More precisely, for all $a_1, \dots, a_n \in N$ we have

$$\mathcal{N} \models \varphi[a_1, \dots, a_n] \iff \mathcal{M} \models \varphi[h(a_1), \dots, h(a_n)].$$

Definition 2.2. A substructure $\mathcal{N}$ of $\mathcal{M}$ is called elementary if the inclusion map is elementary. i.e.

$$\mathcal{N} \models \varphi[a_1, \dots, a_n] \iff \mathcal{M} \models \varphi[a_1, \dots, a_n]$$

for $a_1, \dots, a_n \in N$.

In this case we write $\mathcal{N} \prec \mathcal{M}$ and $\mathcal{N}$ is an elementary substructure of $\mathcal{M}$

Definition 2.3. Let N be a set, now

$$L(N) = L \cup N$$

is the language L with elements of N treated as constants.

Theorem 2.4 (Tarski's Test)**.** *Let $\mathcal{M}$ be an L-structure and $\mathcal{N} \subset \mathcal{M}$. Then $\mathcal{N}$ is an elementary substructure of $\mathcal{M}$ if and only if every $L(N)$-formula $\varphi(x)$ satisfies $\mathcal{M} \models \exists x \varphi(x) \implies \exists a \in N$ s.t. $\mathcal{M} \models \varphi(a)$.*

Example 2.5. Let $L = \{+, 0\}$, $M = (\mathbb{Z})$, $N = 2\mathbb{Z}$ and $\varphi(x) = x + x \doteq 1$. Clearly $\varphi(1)$ holds in $\mathcal{M}$ but there is no $a \in 2\mathbb{Z}$ that satisfies $\varphi(a)$.

Example 2.6. Let $L = \{<\}$, $M = \mathbb{R}$, $N = \mathbb{Q}$ and $\varphi(x) = x < 5$. Now $\varphi(\pi)$ holds in $\mathcal{M}$ but also $\varphi(4.5)$ holds in $\mathcal{N}$. Its clear that actually for any $a \in \mathbb{Q}$ $\varphi(x) = x < a$ and $\varphi(x) = a < x$ can be satisfied. Building up complexity gives statements like $\varphi(x) = (x < a) \wedge (b < x)$ which are clearly satisfiable. (remark: if $N = \mathbb{N}$ then $\varphi(x) = (x < 2) \wedge (1 < x)$ is not satisfiable.) The reader can easily verify, perhaps by induction on complexity, that all $L(N)$ sentences are satisfiable.

Proof. of Tarski's Test. If $\mathcal{N} \prec \mathcal{M}$ and $\mathcal{M} \models \exists x \varphi(x)$, we also have $\varphi(N) \models \exists x \varphi(x)$ and this means there exists $a \in N$ such that $\mathcal{N} \models \varphi(a)$. Thus $\mathcal{M} \models \varphi(a)$.

Now for the other direction, suppose that the condition of Tarski's test is satisfied. Let $f \in \mathcal{L}$ an n-arity function symbol, $a_1, \dots, a_n \in \mathbb{N}$ and $\varphi(x) = f(a_1, \dots, a_n) \doteq x$. Clearly f is closed under N as $\exists a \in N$ such that $\varphi(a)$ is satisfied.

We proceed by induction on the complexity of ψ, that $\mathcal{N} \models \psi \iff \mathcal{M} \models \psi$ for all $L(A)$ sentences ψ. The base case holds trivially from the closure of f. Its also trivial for $\psi = \neg\varphi_1$ and $\psi = \varphi_1 \wedge \varphi_2$.

For $\psi = \exists \varphi_1(x)$.

- If ψ holds in $\mathcal{N}$, there exists $a \in N$ such that $\mathcal{N} \models \varphi_1(a)$. The induction hypothesis yields $\mathcal{M} \models \varphi_1(a)$ thus $\mathcal{M} \models \psi$.
- If ψ holds in $\mathcal{M}$. Then $\varphi_1(x)$ is satisfiable in $\mathcal{M}$, now $\exists a \in N$ such that $\mathcal{M} \models \varphi_1(a)$ and by inductive hypothesis $\mathcal{N} \models \varphi_1(a)$ and $\mathcal{N} \models \psi$ holds.

□

Definition 2.7. $(I, \le)$ is a directed partial order if for all $i, j \in I$ there exists a $k \in I$ such that $i \le k$ and $j \le k$. A family of $(\mathcal{M}_i)_{i\in I}$ of L structures is called directed if

$$i \le j \implies \mathcal{M}_i \subset \mathcal{M}_j.$$

Lemma 2.8. *Let $(\mathcal{M}_i)_{i\in I}$ be a directed family of L-structures. Then $M = \bigcup_{i\in I} M_i$ is the universe of a uniquely determined L-structure $\mathcal{M} = \bigcup_{i\in I} \mathcal{M}_i$.*

Proof. Let R be an n-ary relation symbol and $a_1, \dots, a_n \in M$. As I is directed, there exists $k \in I$ such that all a_i are in A_k. We define (and this is the only possibility)

$$R^{\mathcal{M}}(a_1, \dots, a_n) \iff R^{\mathcal{M}_k}(a_1, \dots, a_n).$$

Constants and function symbols are treated in the same manner. □

Theorem 2.9 (Tarski's Chain Lemma)**.** *The union of an elementary directed family is an elementary extension of all its members.*

Proof. Let $\mathcal{M} = \bigcup_{i\in I}(\mathcal{M}_i)$, we will prove by induction on the complexity of $\varphi(x)$ that for all i and $a \in \mathcal{M}_i$, we have $\mathcal{M}_i \models \varphi(a) \iff \mathcal{M} \models \varphi(a)$.

If φ is atomic

- If $\mathcal{M}_i \models \varphi(a)$ we have $a \in \mathcal{M}_i \subset \mathcal{M}$ so $\mathcal{M} \models \varphi(a)$.
- If $\mathcal{M} \models \varphi(a)$ trivially $\mathcal{M}_i \models \varphi(a)$.

If $\varphi = \varphi_1 \wedge \varphi_2$ then as $\mathcal{M}_i \models \varphi_1(a) \iff \mathcal{M} \models \varphi_1(a)$ and $\mathcal{M}_i \models \varphi_2(a) \iff \mathcal{M} \models \varphi_2(a)$ we have $\mathcal{M}_i \models \varphi_1 \wedge \varphi_2(a) \iff \mathcal{M} \models \varphi_1 \wedge \varphi_2(a)$. Similarly for negation.

If φ is existential, $\varphi(x) = \exists y \psi(x, y)$, then $\varphi(a)$ holds in $\mathcal{M}$ exactly if there exists $b \in M$ with $\mathcal{M} \models \psi(a, b)$. As the family is directed, there always exists a $j \ge i$ with $b \in M_j$. By the induction hypothesis we have $\mathcal{M} \models \psi(a, b) \iff \mathcal{M}_i \models \psi(a, b)$. Thus $\varphi(a)$ holds in $\mathcal{M}$ exactly if it holds in an $\mathcal{M}_j$ $(j \ge i)$. Now claim follows from $\mathcal{M}_i \prec \mathcal{M}_j$. □

Definition 2.10. We call a theory T finitely satisfiable if every finite subset of T is consistent.

Theorem 2.11 (Compactness Theorem)**.** *Finite satisfiable theories are consistent.*

Remark. Here we use Gödel's Completeness Theorem, but its also provable with ultra-products.

Proof. Suppose T is a consistent theory, now T has a model $\mathcal{M}$, now trivially $\mathcal{M}$ satisfies $T' \subset T$ for $|T'| < \aleph_0$.

Now conversely assume that T is inconsistent. Then every sentence φ in L is provable by T. Let φ be any sentence in T. Then there is a proof of $\neg\varphi$ from T, $\varphi'_1, \dots, \varphi'_n = \neg\varphi$. Now let $S = T \cup \{\varphi, \varphi'_1, \dots, \varphi'_n\}$. Then S is a subset of T and both φ and $\neg\varphi$ are provable from it. To see that $\neg\varphi$ is provable from S, observe that φ'_i used in the proof are each either a sentence in T or a consequence of such or a formula that is tautologically true. If S is empty, that is, none of them are in T, then $\neg\varphi$ is provable without T and the claim holds. If there are any $\varphi'_i \in S$ then all of them are axioms of T so that by definition, $\neg\varphi$ is provable from S. Now S is a finite subset such that $S \vdash \varphi$ and $S \vdash \neg\varphi$ that is, S is an inconsistent theory and hence does not have a model. □

Example 2.12 (Robinson's Principal)**.** If a first order sentence holds in every field of characteristic zero, then there exists a constant k such that the sentence holds for every field of characteristic larger than k.

Proof. Suppose φ is a sentence that holds in every field of characteristic zero. Then its negation $\neg\varphi$, together with field axioms and the infinite sequence of sentences

$$1 + 1 \neq 0, 1 + 1 + 1 \neq 0, \dots$$

is not satisfiable. Therefore a finite subset A if these sentences is not satisfiable. A must contain $\neg\varphi$ because otherwise it would be satisfiable. Because adding more sentences to A does not change unsatisfiability, we can assume A contains field axioms and, for some k, the first k sentences of the form $1 + 1 + \ldots + 1 \neq 0$. Let B contain all of the sentences of A other than $\neg\varphi$. Then any field with a characteristic greater than k is a model of B, and $\neg\varphi$ together with B is not satisfiable. This means φ must hold in every model of B, which means φ holds in each field of characteristic greater than k. $\square$

Model Theory Week 4 - Revisiting and applying set theory

June 1, 2023

1 ZFC axioms

ZFC is the "standard" set theory. It includes the original Zermelo-Fraenkel set axioms and the famous axiom of choice (an axiom that brought tons of debates at that time by having some counter-intuitive consequences).

In the context of set theory, elements of sets are also sets. For our language, we will use the first-order logic symbols and the binary relation symbol $\in$. We will also use the following symbols:

$\emptyset$: The constant that represents the empty set.

$\subset$: We say $y \subset x$ (y is contained in x) if $w \in y \to w \in x$.

$s(x)$: We define the successor of a set x as the set $s(x) := x \cup \{x\}$.

f: A function symbol.

The axioms will show that the successor function and the empty set can be defined in set theory. The notion of a function can also be defined using the axioms. Following are the axioms for ZFC:

Extensionality: $(\forall x, y)(x = y \leftrightarrow (\forall z)(z \in x \leftrightarrow z \in y))$
In other words, two sets are the same if they have the same elements.

Null set: $(\exists x \forall y)(\neg y \in x)$
That is, there is a set that contains no other set, also known as the empty set ($\emptyset$).

Pairs: $(\forall x, y \exists z \forall u)(u \in z \leftrightarrow u = x \vee u = y)$
That is, given two sets there exists a set that contains only the two of them, $\{x, y\}$.

Unions: $(\forall x \exists y \forall z)(z \in y \leftrightarrow (\exists w)(z \in w \wedge w \in x))$
Given a set x, there exists a set y that is the union of all sets $y \in x$. This set is denoted as $\bigcup x$.

Power set: $(\forall x \exists y \forall z)(z \in y \leftrightarrow (\exists w)(z \in w \to w \in x))$
Equivalently, $(\forall x \exists y \forall z)(z \subset x \to z \in y)$
Given a set x, there exists the power set of this set, $\mathcal{P}(x)$.

Infinity: $(\exists x)((\exists y)(y \in x) \wedge (\forall y)(y \in x \to (\exists z)(y \in z \wedge z \in x)))$
Equivalently, $(\exists x)((\emptyset \in x) \wedge (\forall y)(y \in x \to s(y) \in x))$
There is a set that is closed under the successor function.

Regularity: $(\forall x)((\exists y)(y \in x) \to (\exists y)(y \in x \wedge \neg(\exists z)(z \in y \wedge z \in x)))$
Every non-empty set is disjoint from one of its elements.

Subsets: $(\forall x \exists y \forall z)(z \in y \leftrightarrow z \in x \wedge \phi(zu_1 \ldots u_n)))$, where ϕ is any formula in which y does not occur.
Intuitively, $y = \{z \in x : \phi(zu_1 \ldots u_n)\}$ is a set. We can describe subsets with formulas.

Collection: $(\forall x)[x \in u \to (\exists z)\phi(xzuv_1 \ldots v_n)] \to (\exists y \forall x)[x \in u \to (\exists z)(z \in y \wedge \phi(xzuv_1 \ldots v_n))]$, where ϕ is a formula in which y does not occur.
If each of the classes $C_x = \{z : \phi(xz \ldots)\}$, $x \in u$, is non-empty, then there exists a set y which meets each of these classes.

Choice $\forall x[\emptyset \in x \to (\exists f : x \to \cup x(\forall y)(f(y) \in y))]$
There exists a choice function that chooses for each element $y \in x$ an element $w \in y$.

2 Ordinals

Definition 2.1 (Totally ordered sets)**.** $\mathcal{L} = \{\leqslant\}$

Reflexive $\forall x,\ x \leqslant x$

Transitive if $x \leqslant y$ and $y \leqslant z$ then $x \leqslant z$

Antisymmetric if $a \leqslant b$ and $b \leqslant a$ then $a = b$

Total $a \leqslant b$ or $b \leqslant a$

Definition 2.2. A totally ordered set x is **well-ordered** if every non-empty subset of x has a least element under the order.

We define the order in relation to $\in$ on a set x as: given $w, y \in x$, $w < y$ if $w \in y$.

Definition 2.3. A set x is an **ordinal** if $x = \emptyset$ or if $\emptyset \in x$ and it is a well-ordered set in relation to the $\in$ order

Theorem 2.4. *A totally ordered set x is well-ordered if, and only if, it is order isomorphic to an ordinal.*

Theorem 2.5 (Well-ordering theorem)**.** *Given x a set, we can give a well-order structure to x.*

Theorem 2.6. *The well-ordering theorem is equivalent to the axiom of choice.*

2.1 Building ordinals

To define the finite ones, let $0 := \emptyset$. We then get recursively the finite sets:

$$1 := s(0) = \{\emptyset\} = \{0\}$$
$$2 := s(1) = \{\emptyset, \{\emptyset\}\} = \{0, 1\}$$
$$3 := s(2) = \{0, 1, 2\}$$
$$\vdots$$

And we can define the total order on these sets as $x \leqslant y$ if $x \in y$ or $x = y$. As these are finite sets and this is always a total order, this is a well-order.

Proposition 2.7. *There exist infinite ordinals.*

Proof. Let
$$\phi(x) := (\emptyset \in x \wedge \forall y(y \in x \to (s(y) \in x)))$$
be a sentence. We want to proof the existence of a unique set ω such that
$$\forall x(x \in \omega \leftrightarrow \forall y(\phi(y) \to x \in I)).$$
Let I be an infinite set, guaranteed by the axiom of infinity. Let
$$W = \{x \in I : \forall J(\phi(J) \to x \in J)\}$$
that is, the minimal element of I satisfying property ϕ. This is our desired set $\omega = \{0, 1, \ldots, n, \ldots\}$. □

One can also define $\omega + 1 := s(\omega) = \{0, 1, 2, \ldots, n, \ldots, \omega\}$ and so on.

Definition 2.8. Given α and β two ordinal, we define the following:

$$\alpha + \beta := \begin{cases} \alpha & \text{if } \beta = 0 \\ (\alpha + \gamma) + 1 & \text{if } \beta = s(\gamma) \\ \cup_{\gamma < \beta}(\alpha + \gamma) & \text{otherwise} \end{cases}$$

$$\alpha \cdot \beta := \begin{cases} 0 & \text{if } \beta = 0 \\ (\alpha \cdot \gamma) + \alpha & \text{if } \beta = s(\gamma) \\ \cup_{\gamma < \beta}(\alpha \cdot \gamma) & \text{otherwise} \end{cases}$$

$$\alpha^\beta := \begin{cases} 1 & \text{if } \beta = 0 \\ (\alpha^\gamma) \cdot \alpha & \text{if } \beta = s(\gamma) \\ \cup_{\gamma < \beta}(\alpha^\gamma) & \text{otherwise} \end{cases}$$

3 Cardinals

Definition 3.1. A cardinal is an ordinal α such that if $\beta \in \alpha$ then there is no one-to-one set function from β to α.

In some way, cardinals are the first ordinals with some size. For example, every element of ω is a cardinal. The set ω itself is a cardinal, but $\omega + 1$ is not.

To differ the notations from ordinals and cardinals, we denote infinite cardinals as $\aleph_\alpha$, for α an ordinal. The first cardinal, ω, is denoted $\aleph_0$.

Definition 3.2. Given x a set, we define the **cardinality** of x as the unique cardinal κ such that there is a set bijection from x to κ. We denote it as $|x|$.

Remark 3.3. $|x| \leqslant |y|$ then there exists and one-to-one set function from x to y and a surjective set function from y to x.

Notice that the well-ordering theorem implies indirectly that every set has a cardinality. If we don't assume the axiom of choice some sets might not have well-defined cardinality in the sense above.

Definition 3.4. Given x, y sets, we define $x^y := \{f : x \to y\}$, the set of all functions with domain x and range y.

Definition 3.5. Let κ_1, κ_2 ordinals and x, y sets with $|x| = \kappa_1$ and $|y| = \kappa_2$. Then:

- $\kappa_1 + \kappa_2 = |x \sqcup y|$, where $\sqcup$ is the disjoint union.
- $\kappa_1.\kappa_2 = |x \times y|$, where $\times$ is the set product.
- $\kappa_1^{\kappa_2} = |x^y|$.

Theorem 3.6. *Let x be a set. Then $|\mathcal{P}(x)| > |x|$. Equivalently, $2^\kappa > \kappa$, for every κ cardinal.*

Proof. Let f be any function from x to $\mathcal{P}(x)$. It suffices to show that f cannot be surjective. Let $T := \{y \in x : y \notin f(x)\}$. By a Cantor diagonal argument it follows that T is not on the image of f, hence f is not surjective. $\square$

The following are two results independent from ZFC:

Conjecture 3.7 (The continuum hypothesis). $2^{\aleph_0} = \aleph_1$.

Conjecture 3.8 (The generalized continuum hypothesis). $2^{\aleph_\alpha} = \aleph_{\alpha+1}$, for every α an ordinal.

4 Consequences of compactness theorem

Theorem 4.1 (Löwenheim–Skolem). *Let $\mathcal{B}$ be an L-structure, S a subset of B and κ an infinite cardinal.*

1. *(Downward) If*
$$\max(|S|, |L|) \leqslant \kappa \leqslant |\mathcal{B}|,$$
then $\mathcal{B}$ has an elementary substructure of cardinality κ containing S.

2. *(Upward) If $\mathcal{B}$ is infinite and*
$$\max(|\mathcal{B}|, |L|) \leqslant \kappa$$
then $\mathcal{B}$ has an elementary extension of cardinality κ.

Proof. (Downward) Choose a set $S \subset S' \subset B$ of cardinality κ and build the minimal elementary substructure of $\mathcal{B}$ containing S'. By (TENT; ZIEGLER, 2012, Crorollary 2.1.3) such substructure exist and it has cardinality restricted at most $\max(|S'|, |L|, \aleph_0)$. By the condition on the cardinality of S' it follows that the cardinality of such substructure is κ.

(Upward) We first construct an elementary extension $\mathcal{B}'$of cardinality at least κ. Choose a set C of new constants such that $|C| = \kappa$. As $\mathcal{B}$ is infinite, the theory

$$\mathrm{Th}(\mathcal{B}_B) \bigcup \{\neg c = f | c, d \in C, c \neq d\}$$

is finitely satisfiable (even in $\mathcal{B}$: just interpret the finitely many new constants in a finite subset by elements of B). By (TENT; ZIEGLER, 2012, Lemma 2.1.1), any model $(\mathcal{B}'_B, b_c)_{c\in C}$ is an elementary extension of $\mathcal{B}$ with κ many different elements $(b_c)_{c\in C}$.

Finally, we apply the downward theorem to $\mathcal{B}'$ and $S = B$. □

Corollary 4.2. *A theory which has an infinite model has a model in every cardinality* $\kappa \geqslant \max(|L|, \aleph_0)$*.*

Definition 4.3. Let κ be an infinite cardinal. A theory T is called **κ-categorical** if all models of T of cardinality κ are isomorphic

Remark 4.4 (Lemma 1.3.9 (TENT; ZIEGLER, 2012))*.* A consistent theory T is complete if, and only if, there exists a structure $\mathcal{U}$ such that $T \equiv \mathrm{Th}(\mathcal{U})$.

Theorem 4.5 (Vaught's test)**.** *A κ-categorical theory T is complete if the following conditions are satisfied:*

a) T is consistent.

b) T has no finite model.

c) $|L| \leqslant \kappa$.

Proof. We have to show that all models $\mathcal{U}$ and $\mathcal{B}$ of T are elementary equivalent. As $\mathcal{U}$ and $\mathcal{B}$ are infinite, $\mathrm{Th}(\mathcal{U})$ and $\mathrm{Th}(\mathcal{B})$ have models $\mathcal{U}'$ and $\mathcal{B}'$ of cardinality κ. By assumption $\mathcal{U}'$ and $\mathcal{B}'$ are isomorphic, and it follows that

$$\mathcal{U} \equiv \mathcal{U}' \equiv \mathcal{B}' \equiv \mathcal{B}.$$

□

5 Classes

The notion of class is not included in ZFC. There are alternative set theories that include classes (Morse–Kelley set theory and NBG, for example). It is still useful and used in other areas of mathematics like category theory.

Definition 5.1. A **class** is a collection of all sets such that a particular condition holds.

For example, ω is the class of all finite ordinals.

A class does not necessarily have to be a set. For example, the class of all sets is not a set (if it was we would have a paradox similar to Russell paradox).

Definition 5.2.

- A class is called a **small class** if it is a set.
- A class is called a **proper class** if it is not a set.

Examples of proper classes are the class of all sets, all groups, and all rings. In general, given T a theory with an infinite model, then the class of all models of T is a proper class.

References

TENT, K.; ZIEGLER, M. *A Course in Model Theory.* [S.l.]: Cambridge University Press, 2012.

1 Historical context

In the late 19th century mathematics was standing before a big problem:

What are we acctually doing?

This is since the foundations of mathematics were very vague until that time. There was some stuff the ancient greeks have written, a lot of handwaving and quite a bit of referencing god to justify mathematical arguments, but no real agreed upon foundation. This functioned suprisingly well for a long time, as long as one only looks at (relatively) finite constructions.

But everything changed when Cantor set forward his set theory, which dealed extensitively with infinite objects. Cantors ambitions of taming infinity could not be realized with the pachwork of intuitions that were based on finite objects. A new firmer foundation had to be created

There were three main approaches to this:

- The Constructivists, rejected all of this. They saw a mathematical object as valid only if it can be constructed. They rejected Cantors set theory, together with any theory that handles objects that are infinite. Additionally they also rejected the priciple of the excluded third and any indirect proof techniques. The most notable contstructivists were Leopold Kronecker and L. E. J. Brouwer.
- The Logisicsts, saw mathematical truth as an extension of logical truth and saw all mathematical objects as logical objects. They thus tried to deduce mathematical facts from purely logical precepcts. The most notable logicists were Bertrand Russell and Gottlob Frege.
- The Constructivists, saw mathematics as a game in which you manipulate symbols with well defined set of rules. Any interpretations of these series of symbols would thus be external to mathematics. The main proponent of Formalism was Hilbert.

These three players all clashed, while trying to develop a firm foundations of mathematics. The most popular formalist foundations of mathematics were the ZFC axioms. Of course the constructivists were not happy with them as they involved a lot of objects that are given without construction. The idea to justify these axioms was to show that they do not lead to contradictions and anything that can be formulated in their languages can be either proven or disproven. It would be nice if it'd worked.

However in 1930 a young mathematician named Kurt Gödel proved that this is impossible as any sufficiently nice logical system that can handle arithmetics cannot be complete. We furthermore cannot even prove that it is cosistent.

2 Proof Theory

We have talked about model theory which looks at theories and families of objects that satisfy this theory. If a sentence is true in each model we call it

provable. There is however another way of attaining provable statments: by proving them!
But what is a proof? Well it is a series of sentences that follow logically from each other. To formalize this notion we will introduce the **predicate calculus**.

Definition 1. For any language $\mathcal{L}$ and any finite collection $\mathfrak{t}$ of sentences and any sentence α we say that α is deriviable from $\mathfrak{t}$ i.e. $\mathfrak{t} \vdash \alpha$ if this can derived from the following rules:

$$\alpha \vdash \alpha$$

$$\vdash t = t$$

$$\frac{\begin{array}{c}\mathfrak{t}_1 \vdash \alpha \\ \mathfrak{t}_2 \vdash \beta\end{array}}{\mathfrak{t}_1 \cup \mathfrak{t}_2 \vdash \alpha \wedge \beta}$$

$$\frac{\mathfrak{t} \vdash (\alpha \wedge \beta)}{\mathfrak{t} \vdash \alpha}$$

$$\frac{\mathfrak{t} \vdash (\alpha \wedge \beta)}{\mathfrak{t} \vdash \beta}$$

$$\frac{\begin{array}{c}\mathfrak{t}_1 \cup \{\alpha\} \vdash \beta \\ \mathfrak{t}_2 \cup \{\neg\alpha\} \vdash \beta\end{array}}{\mathfrak{t}_1 \cup \mathfrak{t}_2 \vdash \beta}$$

$$\frac{\begin{array}{c}\mathfrak{t}_1 \vdash \alpha \\ \mathfrak{t}_2 \vdash \beta\end{array}}{\mathfrak{t}_1 \cup \mathfrak{t}_2 \vdash \beta}$$

$$\frac{\mathfrak{t} \vdash \forall x,\ \alpha}{\mathfrak{t} \vdash \alpha}$$

$$\frac{\mathfrak{t} \vdash \alpha}{\mathfrak{t} \vdash \forall x,\ \alpha}$$

This rule holds only for x that do not appear free in in any expression in $\mathfrak{t}$

$$\frac{\mathfrak{t} \vdash \alpha}{\mathfrak{t}|_{x=t} \vdash \alpha|_{x=t}}$$

$$\frac{\mathfrak{t} \vdash \alpha}{\mathfrak{t} \cup \{x = t\} \vdash \alpha|_{x=t}}$$

For a potentially infinite collection of expressions $\mathfrak{T}$ we have $\mathfrak{T} \vdash \alpha$ if there is some finite collection of expressions $\mathfrak{t}$ contained in $\mathfrak{T}$, s.t. $\mathfrak{t} \vdash \alpha$

This definition alows us to very derive statements from a theory via rules that seem logical. Now we can say that a collection of expressions $\mathfrak{T}$ is consistent if we have for any (or some) expression α

$$\mathfrak{T} \nvdash (\alpha \wedge \neg\alpha)$$

We will call it complete if for any expression α:

$$\mathfrak{T} \vdash \alpha$$

or:

$$\mathfrak{T} \vdash \neg\alpha$$

Now we can ask ourself whether this definition of provability from model theory (i.e. being true in each model) is the same as the one from proof theory. It is pretty easy to check that the provability in the theory implies provability in a model.

Theorem 1. *For any set of expressions $\mathfrak{T}$ and any expression α we have:*

$$\mathfrak{T} \vdash \alpha \implies \mathfrak{T} \vDash \alpha$$

Proof. We just have to check each step of the derivation rules whether if the presuppositions hold in the model the the conclusion also holds.

□

In order to prove the converse we will have to show the Gödel completeness theorem which shows that any theory has a model:

Theorem 2. *Any consistent set of expressions $\mathfrak{T}$ is satisfiable (has an interpretation).*

Proof. We will first assume that there are infinitely many variables that do not appear in any expression of $\mathfrak{T}$ (this can be accomplished by a simple shift). We will expand the theory recursively to finally get a nice maximal theory. To do this we will enumarate all terms $t_0, t_1, t_2, \dots$ and all expressions $\alpha_0, \alpha_1, \alpha_2, \dots$. Using this we will recursively define $\mathfrak{T}_i$ for any natural number i.
First we can define $\mathfrak{T}_0 := \mathfrak{T}$
For this let $y_0, y_1, y_2, \dots$ be the infinitely many distinct variables that do not appear in $\mathfrak{T}_i$, nor in α_i, nor in t_i. Let $\mathfrak{S}_i$ be the collection of the expressions of the form $y_{2k} = t_i$ for any natural number k.
We will first look at the case where $\mathfrak{T}_i \cup \{\alpha_i\}$ is consistent. In this case we can simply define:

$$\mathfrak{T}_{i+1} := \mathfrak{T}_i \cup \mathfrak{S}_i \cup \{\alpha_i\}$$

Next if $\mathfrak{T}_i \cup \{\alpha_i\}$ is inconsistent. We look at whether α_i is a generalisation (i.e. $\alpha_i \equiv \forall x \gamma_i$ for some γ_i). If α_i is not a generalization we simply define:

$$\mathfrak{T}_{i+1} := \mathfrak{T}_i \cup \mathfrak{S}_i$$

If $\alpha_i \equiv \forall x \gamma_i$ for some γ_i, note that y_1 does not appear in γ_i and thus we can define $\beta_i := \gamma_i|_{x=y_0}$. Then we define:

$$\mathfrak{T}_{i+1} := \mathfrak{T}_i \cup \mathfrak{S}_i \cup \{\neg\beta_i\}$$

We note that the variables $y_3, y_5, y_7, \dots$ do not appear in $\mathfrak{T}_{i+1}$ so we can iterate the construction.
Now let $\mathfrak{T}^*$ be the union of all $\mathfrak{T}_i$ for $i \geq 0$. We can see that since each $\mathfrak{T}_i$ is consistent and for any i we have $\mathfrak{T}_i \subseteq \mathfrak{T}_{i+1}$, $\mathfrak{T}^*$ must also be consistent.
Furthermore we can also note that:

$$\mathfrak{T}^* \vdash \alpha \iff \alpha \in \mathfrak{T}^*$$

Now we can use the construction of $\mathfrak{T}^*$ to construct an interpretation for it. This will also give us an interpretation for $\mathfrak{T}$ since it is contained in $\mathfrak{T}^*$.
For the model of $\mathfrak{T}^*$ we look at the collection of all terms in the language: τ and introduce the equivalence relation $\sim$ on it by setting:

$$t \sim t' \iff t = t' \in \mathfrak{T}^*$$

Then the collection of individuals that we will form an interpretation over will be $\omega := \tau/\sim$. Let $\bar{t}$ be the equivalence class of some term t.
We can then define the interpretation $\mathfrak{I}$ over ω as follows:

$$\begin{aligned}
\mathfrak{I}(C) &:= \bar{C} \\
\mathfrak{I}(x) &:= \bar{x} \\
\mathfrak{I}(f)(\bar{t_1}, \dots, \bar{t_r}) &:= \overline{f(t_1, \dots, t_r)} \\
\mathfrak{I}(R)(\bar{t_1}, \dots, \bar{t_r}) &\iff R(t_1, \dots, t_n)
\end{aligned}$$

Were C is a constant symbol, x is a variable, f is a function symbol and R is a relation symbol in $\mathcal{L}$ (with r being the arity of the function and relation).
Now to show that this interpretation actually satisfies every sentence in $\mathfrak{T}^*$ we can proceed by induction over the rank of a sentence $\alpha \in \mathfrak{T}^*$. □

Now this shows the converse of the previous theorem we can simply note that if $\mathfrak{T} \vDash \alpha$ then we have that in each model of $\mathfrak{T}$ the proposition α holds, but since each model of $\mathfrak{T} \cup \{\neg\alpha\}$ also is a model of $\mathfrak{T}$, the sentence α holds in it, but since $\neg\alpha$ also holds this is a contradiction. So $\mathfrak{T} \cup \{\neg\alpha\}$ has no model and thus cannot be consistent and thus we can prove anything from it giving us $\mathfrak{T} \cup \{\neg\alpha\} \vdash \alpha$. But since obviously $\mathfrak{T} \cup \{\alpha\} \vdash \alpha$ we must have $\mathfrak{T} \vdash \alpha$.

3 Incompleteness

But as Gödel giveth, Gödel taketh away. While we have a connection between proofs and model we can also show that in most cases we cannot have a theory that is complete (under some weak assumptions) and we cannot show the consistency of a system within the system. What Hilbert and co. hoped to do is find a universal perfect system where you could encode all of mathematics. This system should of course be consistent and ideally complete, but then Gödel came and showed that as long as the system can do arithmetic (and is consistent) it is impossible for it to be complete and even worse we cannot even show that it's consistent.
To do this we embed a basic system $\mathfrak{S}$ of arithmetic into our big system $\mathfrak{T}$ and introduce an encoding for any formula in the language of $\mathfrak{T}$ as numbers and thus we will be able to handle them in $\mathfrak{S}$.
We will denote the code for a formula ϕ by $\lceil\phi\rceil$. We can for the coding define function symbols neg and imp s.t.

$$\mathfrak{S} \vdash \mathrm{neg}(\lceil\phi\rceil) = \lceil\neg\phi\rceil$$
$$\mathfrak{S} \vdash \mathrm{imp}(\lceil\phi\rceil, \lceil\psi\rceil) = \lceil\phi \to \psi\rceil$$

furthermore for a formula ϕx where x is a free variable we can define for any term t the function symbol sub s.t.

$$\mathfrak{S} \vdash \mathrm{sub}(\lceil\phi x\rceil, \lceil t\rceil) = \lceil\phi t\rceil$$

If we have such a coding we can additionally code any sequences of formulas Φ as a number, denoting their code again as $\lceil\Phi\rceil$. Now we can also see in $\mathfrak{S}$ whether the sequence of formulas is a proof and what their last formula. So we can define the sentence in $\mathfrak{S}$: $\mathrm{Prov}(\lceil\Phi\rceil, \lceil\phi\rceil)$ s.t. $\mathfrak{S} \vdash \mathrm{Prov}(\lceil\Phi\rceil, \lceil\phi\rceil)$ iff Φ is a proof of ϕ in $\mathfrak{T}$. Furthermore we can define Pr by:

$$\mathrm{Pr}(y) \equiv \exists x \mathrm{Prov}(x, y)$$

Now it is clear that:

$$\mathfrak{T} \vdash \phi \implies \mathfrak{S} \vdash \mathrm{Pr}(\lceil\phi\rceil) \tag{1}$$

additionally we will require:

$$\mathfrak{S} \vdash \mathrm{Pr}(\ulcorner\phi\urcorner) \to \mathrm{Pr}(\ulcorner\mathrm{Pr}(\ulcorner\phi\urcorner)\urcorner) \tag{2}$$

$$\mathfrak{S} \vdash \mathrm{Pr}(\ulcorner\phi\urcorner) \wedge \mathrm{Pr}(\ulcorner\phi \to \psi\urcorner) \to \mathrm{Pr}(\ulcorner\psi\urcorner) \tag{3}$$

To show the incompleteness theorem we have to further demand of our theory $\mathfrak{T}$ a weakened version of the converse of 1:

$$\mathfrak{T} \vdash \mathrm{Pr}(\ulcorner\phi\urcorner) \implies \mathfrak{T} \vdash \phi \tag{4}$$

this is not true for every $\mathfrak{T}$, but any theory without such a property produces obviously untrue statements.
Now if we want to find a formula that is not provable in $\mathfrak{T}$ we will first have to prove the diagonalization lemma:

Lemma 1. *For any sentence ϕ in the language of $\mathfrak{T}$ that has one free variable we have some formula ψ s.t.*

$$\mathfrak{S} \vdash \psi \leftrightarrow \phi(\ulcorner\psi\urcorner)$$

Proof. We first define $\theta(x) \equiv \phi(\mathrm{sub}(x,x))$, $m \coloneqq \ulcorner\theta\urcorner$ and $\psi \equiv \theta(m)$. Then we have in $\mathfrak{S}$

$$\begin{aligned} \psi &\leftrightarrow \theta(m) && (5)\\ &\leftrightarrow \phi(\mathrm{sub}(m,m)) && (6)\\ &\leftrightarrow \phi(\mathrm{sub}(\ulcorner\theta\urcorner,m)) && (7)\\ &\leftrightarrow \phi(\ulcorner\theta(m)\urcorner) && (8)\\ &\leftrightarrow \phi(\ulcorner\psi\urcorner) && (9) \end{aligned}$$

□

Now if we plug in $\neg\mathrm{Pr}$ as ϕ we get the unprovable statement:

Theorem 3. *Let ψ be s.t.*

$$\mathfrak{S} \vdash \psi \leftrightarrow \neg\mathrm{Pr}(\ulcorner\psi\urcorner)$$

then we have $\mathfrak{T} \nvdash \psi$ and $\mathfrak{T} \nvdash \neg\psi$

Proof. To show that $\mathfrak{T} \nvdash \psi$ we note that if $\mathfrak{T} \vdash \psi$ we must have by 1 $\mathfrak{T} \vdash \mathrm{Pr}(\ulcorner\psi\urcorner)$ i.e. $\mathfrak{T} \vdash \neg\psi$ which contradicts $\mathfrak{T}$ being consistent.
Now if $\mathfrak{T} \vdash \neg\psi$ we have $\mathfrak{T} \vdash \mathrm{Pr}(\ulcorner\psi\urcorner)$ but then by our additional assumption 4 we must have $\mathfrak{T} \vdash \psi$ which contradicts $\mathfrak{T}$ being consistent. □

Model Theory Week 6 - FA and QFA groups

March 30, 2023

1 Main ideas and first definition

The strongest property of a group G is that of 'being isomorphic to G'. If we can describe this by a unique first-order sentence, G is said to be finitely axiomatizable (FA).

For example, let $(*,^{-1},e)$ our language of groups. Then

$$(\exists x)((x \neq e) \wedge (\forall z)((z \neq e) \rightarrow (z = x)))$$

is the first order-sentence describing the cyclic group of order 2, that is, C_2 is (FA). It is also the case that any finite group is (FA), where they can be described by stating that the group has exactly n elements and writing down the rules from the multiplication table. On the other hand, infinite groups cannot be (FA) by the Löwenheim-Skolem theorem, so to make our question interesting, we need to restrict to limit the groups we consider.

Definition 1.1. A finitely generated, infinite group G is **QFA** (for quasi-finitely axiomatizable) if there exists a first-order sentence ϕ such that if H is a finitely generated group and $H \models \phi$ then $H \cong G$.

Instead of working on the entire class of groups, we restrict our definition to the finitely-generated groups.

A result about profinite groups shows that if G is a (topologically) finitely generated profinite group, then $Th(G)$ characterizes G up to isomorphism among all profinite groups (also known as being quasi-axiomatizable). This is not true for abstract, finitely generated groups, as, for example, all non-abelian free groups have the same theory. We will focus on the question of which groups are QFA in the class of finitely generated profinite groups.

Definition 1.2. Let L be a language containing the language of groups and $\mathcal{C}$ a class of groups. We say that a group G is FA (with respect to L) in $\mathcal{C}$ if G is contained in the class $\mathcal{C}$ and there exist a sentence σ_G of L such that for any group H in $\mathcal{C}$,

$$H \models \sigma_G \text{ if and only if } H \cong G.$$

In this case, QFA means FA restricted to the class of finitely generated groups. When $\mathcal{C}$ is a class of profinite groups, isomorphisms are required to be topological.

2 Results from other papers that are central

Given G a group, denote $Z(G)$ the centre of G and $\Delta(G)/G'$ the torsion subgroup of G/G', where G' is the derived group.

Theorem 2.1. *Let G be a group such that $Z(G) \nsubseteq \Delta(G)$. If ϕ is a sentence such that $G \models \phi$, then $G \times C_p \models \phi$ for almost all primes p.*

Proposition 2.2. *Let $\{G_i\}_{i \in I}$ be an infinite family of groups. If ϕ is a sentence such that $\prod_{i \in I} G_i \models \phi$, then there exists $q \neq r \in I$ such that*

$$G_r \times \prod_{i \in I \setminus \{q\}} G_i \models \phi.$$

Proposition 2.3. *1. If the pronilpotent (inverse limit of nilpotent) group G is FA in the class of profinite, or pronilpotent, group, then the set of primes p such that G has a nontrivial Sylow prop-p subgroup is finite.*

2. Let $\mathcal{G}$ be a linear algebraic group of positive dimension defined over $\mathbb{Q}$. If $\mathcal{G}(\mathbb{Z}_\pi)$ is FA in the class of profinite groups, then the set of primes π is finite.

Here $\mathbb{Z}_\pi := \prod_{p\in\pi} \mathbb{Z}_p$.

3 Definable subgroups

Definition 3.1. For a group G and a formla $\kappa(x)$ (possibly with parameters $\overline{g}$ from G), we write

$$\kappa(G) = \kappa(\overline{g}; G) := \{x \in G | G \models \kappa(\overline{g}, x)\}.$$

A subgroup is definable if it is of this form.

Unless otherwise stated, κ is supposed to be a formula of L_{gp}. Note that $\kappa(G)$ is a subgroup if and only if $G \models s(\kappa)$ where

$$s(\kappa) \equiv ((\exists x)\kappa(x) \wedge (\forall y, z)(\kappa(y) \wedge \kappa(z) \to \kappa(y^{-1}z)))$$

and $\kappa(G)$ is a normal subgroup if and only if $G \models s_\triangleleft(\kappa)$ where

$$s_\triangleleft(\kappa) \equiv (s(\kappa) \wedge (\forall x, y)(\kappa(x) \to \kappa(y^{-1}xy))).$$

We will say that a subgroup H is definably closed if $H = \kappa(G)$ for a formula κ such that in any profinite group M, the subset $\kappa(M)$ is necessarily closed.

Theorem 3.2. *Let G be a finitely generated profinite group.*

1. *Every subgroup of finite index is both open and definably closed (with parameters).*
2. *Each term $\gamma_n(G)$ of the lower central series of G is closed and definable (without parameters).*
3. *Every group homomorphisms from G to a profinite group is continuous.*

Definition 3.3. If there is a finite (usually generating) tuple $\overline{g}$ in G, there is a formula σ_G such that for a group H in $\mathcal{C}$ and a tuple $\overline{h}$ in H, $H \models \sigma_G(\overline{h})$ if and only if there is an isomorphism from G to H mapping $\overline{g}$ to $\overline{h}$. In this case we say $(G, \overline{g})$ is FA in $\mathcal{C}$.

Definition 3.4. Given a group N and elements $h_1, \ldots, h_s \in N$, we say that an element g of N is $\overline{h}$-definable in N if there is a formula ϕ_g such that for $c \in N$,

$$N \models \phi_g(\overline{h}, c) \Leftrightarrow c = g.$$

This holds in particular if $g \in \langle h_1, \ldots, h_s\rangle$.

Theorem 3.5. *Let*

$$N = \overline{\langle h_1, \ldots, h_s\rangle} \triangleleft_o G = \overline{\langle g_1, \ldots, g_r\rangle} \in \mathcal{C}$$

and assume that $(N, \overline{h})$ is FA. Then G is FA provided one of the following holds:

(a) $N \cap \langle g_1, \ldots, g_r\rangle = \langle h_1, \ldots, h_s\rangle$, in which case $(G, \overline{g})$ is FA; or

(b) $Z(N) = 1$, $\{h_1, \ldots, h_s\} \subseteq \langle g_1, \ldots, g_r\rangle$ and $h_i^{g_j}$ is $\overline{h}$-definable in N for each i and j.

4 Bi-interpretanbility

All rings are commutative, with identity. L_{rg} is the first-order language of rings. The rings R is **additively profinite** if its additive group $(R, +)$ is profinite as a group.

Definition 4.1. A profinite ring R is **FA (respectively, strongly FA)** if there is a formula σ of L_{rg} such that (i) $R \models \sigma$ and (ii) if S is a profinite (respectively, additively profinite) ring and $S \models \sigma$, then S is topologically isomorphic to R.

Definition 4.2. Let R be an additively profinite ring. We say that R is topologically interpreted in a profinite group G if there are formulae τ, μ and a tuple of parameters $\overline{g}$ in G with the following property:

- for every profinite group H and tuple $\overline{h}$ from H, the set $\tau(\overline{h}; H)$ is a closed subgroup of H;
- $\tau(\overline{g}; G)$ becomes a ring $\hat{\tau}(G) = \hat{\tau}(\overline{g}; G)$ with ring addition given by the group operation, and ring multiplication defined by $res(\mu(\overline{g}), \tau(\overline{g}))$, in the sense that for $x, y, z \in \tau(\overline{g}; G)$,

$$x.y = z \Leftrightarrow G \models \mu(\overline{g}, x, y, z).$$

- $\hat{\tau}(\overline{g}; G)$ is topologically isomorphic to R.

Definition 4.3. Let G be a profinite group. We say that G is topologically interpreted in a profinite ring R if for some d natural number, there are L_{rg} formulae α_1, α_2 such that

- for every profinite ring T, the subset $\alpha_1(T^{(d)})$ is closed in $T^{(d)}$;
- $\alpha_1(R^{(d)})$ is a group $\hat{\alpha}(R)$, with operation defined by

$$\overline{a}.\overline{b} = \overline{c} \Leftrightarrow R \models \alpha_2(\overline{a}, \overline{b}, \overline{c});$$

- G is topologically isomorphic to $\hat{\alpha}(R)$ (with the subspace topology induced by $\overline{\alpha}(R) \subseteq R^{(d)}$).

Definition 4.4. Let G be a profinite group and R a profinite ring. We say that G and R are topologically bi-interpretable in the following circumstances:

- R is topologically interpreted in G by τ and G is topologically interpreted in R by $\overline{\alpha}$, as above;
- identifyin G with $\hat{\alpha}(R) \subseteq R^{(d)}$ and R with $\hat{\tau}(G) \subset G$ gives two mappings

$$\theta : G = \hat{\alpha}(R) \to \hat{\alpha}(\hat{\tau}(G)) \hookrightarrow G^{(d)}$$

$$\theta' : R = \hat{\tau}(G) \to \hat{\tau}(\hat{\alpha}(R)) \hookrightarrow R^{(d)}$$

  then θ is L_{gp}-definable and θ' is L_{rg}-definable.

Let R be a ring. The group

$$Af_1(R) = \begin{pmatrix} 1 & R \\ 0 & R^* \end{pmatrix} < GL_2(R).$$

and R are bi-interpretable. More than that, if R is FA, then the group $Af_1(R)$ is FA in profinite groups.

Definitions

Definition 4.5. Let G be a group. Define the lower central series of G as:

$$\gamma_0(G) := G,$$

$$\gamma_{i+1}(G) = [G, \gamma_i(G)].$$

If there is $n \in \mathbb{N}$ such that $\gamma_n(G) = 1$ then G is said to be nilpotent.

Model Theory

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7 Definable Sets

Definition 7.1. Let $\mathcal{M}$ be an L-structure with universe M. We say that $X \subset M^n$ is *definable* if and only if there is an L-formula $\phi(v_1, \ldots, v_n, w_1, \ldots, w_m)$ and $\bar{b} \in M^m$ such that $X = \{\bar{a} \in M^n : \mathcal{M} \models \phi(\bar{a}, \bar{b})\}$. We say that $\phi(\bar{v}, \bar{b})$ *defines* X. We say that X is *A-definable* or *definable over A* if there is a formula $\psi(\bar{v}, w_1, \ldots, w_l)$ and $\bar{b} \in A^l$ such that $\psi(\bar{v}, \bar{b})$ defines X.

Example 7.2. Let $\mathcal{M} = (\mathbb{Z}, +, -, \cdot, 0, 1)$ be the ring of integers and let $X = \{(m, n) \subset \mathbb{Z}^2, m < n\}$. Then, X is definable (indeed $\emptyset$-definable). By Lagrange's theorem, every non-negative integer is the sum of four squares. Thus, if we let $\phi(x, y)$ be the formula

$$\exists z_1 \exists z_2 \exists z_3 \exists z_4 (z_1 \not\doteq 0 \wedge y \doteq x + z_1^2 + z_2^2 + z_3^2 + z_4^2),$$

then $X = \{(m, n) \in \mathbb{Z}^2; \mathcal{M} \models \phi(m, n)\}$.

Example 7.3. Let F be a field and $\mathcal{M} = (F[X], +, -, \cdot, 0, 1)$ be the ring of polynomials over F. Then F is definable in $\mathcal{M}$. Indeed, F is the set of units of $F[X]$ and is defined by the formula $x \doteq 0 \vee \exists y\ xy = 1$

Proposition 7.4. *Let L_r be the language of ordered rings and $(\mathbb{R}, +, -, \cdot, <, 0, 1)$ be the ordered field of real numbers. Suppose that $X \subset \mathbb{R}^n$ is A-definable. Then, the topological closure of X is also A-definable.*

Proof. Let $\phi(\bar{v}, \bar{a})$ define X. Let $\psi(\bar{v}, \bar{w})$ be the formula

$$\forall \varepsilon \left[\varepsilon > 0 \rightarrow \exists \bar{y} (\phi(\bar{y}, \bar{w}) \wedge \|\bar{v} - \bar{y}\| < \varepsilon)\right].$$

Then $\bar{b}$ is in the closure of X if and only if $\mathcal{M} \models \phi(\bar{b}, \bar{a})$. □

Definition 7.5. A L-term is either

1. a constant symbol c
2. a variable symbol v_i
3. a function applied to a tuple of terms $f(t_1, \ldots, t_{n_f})$

Proposition 7.6 (The Characterisation of Definable Sets)**.** *Let $\mathcal{M}$ be an L-structure. Suppose that D_n is a collection of subsets of M^n for all $n \geq 1$ and $\mathcal{D} = (D_n : n \geq 1)$ is the smallest collection such that:*

i) $M^n \in D_n$

ii) for all n-ary function symbols f of L, the graph of $f^{\mathcal{M}}$ is in D_{n+1}

iii) for all n-ary relation symbols R of L, $R^{\mathcal{M}} \in D_n$

iv) for all $i, j \leq n$, $\{(x_1, \ldots, x_n) \in M^n; x_i = x_j\} \in D_n$

v) if $X \in D_n$, then $M \times X \in D_{n+1}$

vi) each D_n is closed under complement, union and intersection

vii) if $X \in D_{n+1}$ and $\pi : M^{n+1} \to M^n$ is the projection map $(x_1, \dots, x_{n+1}) \mapsto (x_1, \dots, x_n)$, then $\pi(X) \in D_n$

viii) if $X \in D_{n+m}$ and $b \in M^m$, then $\{a \in M^n : (a, b) \in X\} \in D_n$

Then, $X \subset M^n$ is definable if and only of $X \in D_n$.

Proof. We first show that the definable sets satisfy the closure properties i)-viii). Because $\mathcal{D}$ is the smallest collection with these properties, every $X \in \mathcal{D}_n$ is definable.

i) M^n is definable by $v_1 = v_1$.

ii) The graph of $f^{\mathcal{M}}$ is definable by $f(x_1, \dots, x_{n_f} = y$.

iii) The relation $R^{\mathcal{M}}$ is defined by R.

iv) The set $\{x \in M^n : x_i = x_j\}$ is definable by $v_i = v_j$.

v) If $X \subset M^n$ is defined by $\phi(v_1, \dots, v_n, \bar{a})$, then $M \times X$ is defined by $\phi(v_2, \dots, v_{n+1}, \bar{a})$.

vi) If $X \subset M^n$ is defined by $\phi(\bar{v}, \bar{a})$ and $Y \subset M^n$ is defined by $\psi(\bar{v}, \bar{b})$, then $M \setminus X$ is defined by $\neg\phi(\bar{v}, \bar{a})$, $X \cap Y$ is defined by $\phi(\bar{v}, \bar{a}) \wedge \psi(\bar{v}, \bar{b})$ and $X \cup Y$ is defined by $\phi(\bar{v}, \bar{a}) \vee \psi(\bar{v}, \bar{b})$.

vii) If $X \subset M^{n+1}$ us defined by $\phi(v_1, \dots, v_{n+1}, \bar{a})$, then $\pi(X)$ is defined by $\exists v_{n+1} \phi(v_1, \dots, v_{n+1}, \bar{a})$.

viii) If $X \subset M^{n+m}$ is defined by $\phi(v_1, \dots, v_{n+m}, \bar{c})$ and $\bar{b} \in M^n$, then $\{\bar{a} \in M^n; (\bar{a}, \bar{b})\}$ is defined by $\phi(v_1, \dots, v_n, \bar{b}, \bar{c})$. Thus, if $X \in D_n$, then X is definable.

Next we show that if $X \subset M^n$ is definable, then $X \in D_n$. We first show by induction that if $t(v_1, \dots, v_n)$ is a term, then $\{(\bar{x}, y) \in M^{n+1}; t^{\mathcal{M}}(\bar{x}) = y\} \in D_{n+1}$.

If t is a constant term c, then we must show that $\{(\bar{x}, c^{\mathcal{M}}); \bar{x} \in M^n\} \in D_{n+1}$. By iv) and viii) $\{c^{\mathcal{M}} \in D_1\}$. Thus, by n applications of v), $\{(\bar{x}, c^{\mathcal{M}}); \bar{x} \in M^n\} \in D_{n+1}$.

If t is v_i, then we must show that $\{(\bar{x}, x_i); \bar{x} \in M^n\} \in D_{n+1}$, but this follows easily from i) and iv).

Suppose that $t = f(t_1, \dots, t_m)$. By induction we suppose that $G_i \in D_{n_i}$, where G_i is the graph $t_i^{\mathcal{M}} : M^n \to M$. Let $G \in D_{m+1}$ be the graph of $f^{\mathcal{M}}$. Then, the graph of $t^{\mathcal{M}}$ is

$$\left\{(\bar{x}, y); \exists z_1 \dots \exists z_m \left(\bigwedge_{i=1}^{m} (\bar{x}, z_i) \in G_i \wedge (\bar{z}, y) \in G\right)\right\}.$$

Using the closure properties of $\mathcal{D}$, we see this is in D_{n+1}.

We now proceed by induction on formulas to show that every $\emptyset$-definable $X \subset M^n$ is in D_n.

Let ϕ be $t_1 \doteq t_2$. Then $\{\bar{x} \in M^n; t_1^{\mathcal{M}}(\bar{x}) = t_2^{\mathcal{M}}(\bar{x})\} = \{\bar{x} \in M^n; \exists y \exists z (t_1^{\mathcal{M}}(\bar{x}) = y \wedge t_2^{\mathcal{M}}(\bar{x}) = z \wedge y = z\} \in D_n$.

Let ϕ be $R(t_1, \dots t_m)$. Then $\{\bar{x} \in M^n; \mathcal{M} \models \phi(\bar{x})\} =$

$$\left\{(\bar{x}, y); \exists z_1 \dots \exists z_m \bigwedge_{i=1}^{m} t_i^{\mathcal{M}}(\bar{x}) = z_i \wedge \bar{z} \in R^{\mathcal{M}}\right\}$$

Thus, all sets defined over $\emptyset$ by atomic formulas are in $\mathcal{D}$. Because $\mathcal{D}$ is closed under boolean combinations and projections, all $\emptyset$-definable sets are in $\mathcal{D}$. Property viii) ensures that all definable sets are in $\mathcal{D}$. □

Proposition 7.7. *Suppose that $\mathcal{M}$ is a substructure of $\mathcal{N}$, $\bar{a} \in M$, and $\phi(\bar{v})$ is a quantifier-free formula. Then, $\mathcal{M} \models \phi(\bar{a})$ if and only if $\mathcal{N} \models \phi(\bar{a})$.*

Proof. idea, induction on complexity. □

Definition 7.8. The set $\mathrm{Diag}(\mathcal{M})$ is the set of all L_M atomic formulas $\phi(m_1, \dots, m_n)$, or negations thereof, such that $\mathcal{M} \models \phi(m_1, \dots, m_n)$.

Lemma 7.9. *Suppose that $\mathcal{N}$ is an L_M-structure and $\mathcal{N} \models (\mathrm{Diag}(\mathcal{M}))$; then, viewing $\mathcal{N}$ as an L-structure, there is an L-embedding of $\mathcal{M}$ into $\mathcal{N}$.*

Definition 7.10. We say that a theory T has *quantifier elimination* if for every formula ϕ there is a quantifier-free formula ψ such that

$$T \models \phi \leftrightarrow \psi.$$

Theorem 7.11. *Suppose that L contains a constant symbol c, T is an L-theorey and $\phi(\bar{v})$ is an L-formula. The following are equivalent:*

i) There is a quantifier-free L-formula $\psi(\bar{v})$ such that $T \models \forall \bar{v}(\phi(\bar{v}) \leftrightarrow \psi(\bar{v}))$.

ii) If $\mathcal{M}$ and $\mathcal{N}$ are models of T, $\mathcal{A}$ is an L-structure, $\mathcal{A} \subset \mathcal{M}$ and $\mathcal{A} \subset \mathcal{N}$, then $\mathcal{M} \models \phi(\bar{a})$ if and only if $\mathcal{N} \models (\bar{a})$ for all $\bar{a} \in \mathcal{A}$.

Proof. i) $\Longrightarrow$ ii) Suppose that $T \models \forall \bar{v}(\phi(\bar{v}) \leftrightarrow \psi(\bar{v}))$, where ψ is quantifier free. Let $\bar{a} \in \mathcal{A}$, where $\mathcal{A}$ is a common substructure of $\mathcal{M}$ and $\mathcal{N}$ and the latter two structures are models of T. In proposition 7.7, we see that quantifier-free formulas are preserved under substructure and extension. Thus

$$\begin{aligned} \mathcal{M} \models \phi(\bar{a}) &\iff \mathcal{M} \models \psi(\bar{a}) \\ &\iff \mathcal{A} \models \psi(\bar{a}) \text{ (because } \mathcal{A} \subset \mathcal{M}) \\ &\iff \mathcal{N} \models \psi(\bar{a}) \text{ (because } \mathcal{A} \subset \mathcal{N}) \\ &\iff \mathcal{N} \models \phi(\bar{a}) \end{aligned}$$

ii) $\Longrightarrow$ i) First, if $T \models \forall \bar{v}\phi(\bar{v})$, then $T \models \forall \bar{v}(\phi(\bar{v}) \leftrightarrow c \doteq c)$. Second, if $T \models \forall \bar{v} \neg \phi(\bar{v})$, then $T \models \forall \bar{v}(\phi(\bar{v}) \leftrightarrow c \neq c)$.

Thus, we may assume that both $T \cup \{\phi(\bar{v})\}$ and $T \cup \{\neg\phi(\bar{v})\}$ are satisfiable.

Let $\Gamma(\bar{v}) = \{\psi(\bar{v}); \psi$ is quantifier-free and $T \models \forall \bar{v}(\phi(\bar{v}) \to \psi(\bar{v}))\}$. Let $d_1, \ldots, d_m$ be new constant symbols. We will show that $T \cup \Gamma(\bar{d}) \models \phi(\bar{d})$. Then, by compactness, there are $\psi_1, \ldots, \psi_n \in \Gamma$ such that

$$T \models \forall \bar{v} \left(\bigwedge_{i=1}^{n} \psi_i(\bar{v}) \to \phi(\bar{v}) \right).$$

Thus

$$T \models \forall \bar{v} \left(\bigwedge_{i=1}^{n} \psi_i(\bar{v}) \leftrightarrow \phi(\bar{v}) \right)$$

and $\bigwedge_{i=1}^{n} \psi_i(\bar{v})$ is quantifier-free. We need only prove that $T \cup \Gamma(\bar{d}) \models \phi(\bar{d})$.

Suppose not. Let $\mathcal{M} \models T \cup \Gamma(\bar{d}) \cup \{\neg\phi(\bar{d})\}$. Let $\mathcal{A}$ be the substructure of $\mathcal{M}$ generated by $\bar{d}$.

Let $\Sigma = T \cup \mathrm{Diag}(\mathcal{A}) \cup \phi(\bar{d})$. If Σ is unsatisfiable, then there are quantifier-free formulas $\psi_1(\bar{d}), \ldots, \psi_n(\bar{d}) \in \mathrm{Diag}(\mathcal{A})$ such that

$$T \models \forall \bar{v} \left(\bigwedge_{i=1}^{n} \psi_i(\bar{v}) \to \neg\phi(\bar{v}) \right).$$

But then

$$T \models \forall \bar{v} \left(\phi(\bar{v}) \to \bigvee_{i=1}^{n} \neg\psi_i(\bar{v}) \right),$$

so $\bigvee_{i=1}^{n} \neg\psi_i(\bar{v}) \in \Gamma$ and $\mathcal{A} \models \bigvee_{i=1}^{n} \neg\psi_i(\bar{d})$, a contradiction. Thus, Σ is satisfiable. Let $\mathcal{N} \models \Sigma$. Then $\mathcal{N} \models \phi(\bar{d})$. Because $\Sigma \supset \mathrm{Diag}(\mathcal{A})$, $\mathcal{A} \subset \mathcal{N}$, by lemma 7.9. But $\mathcal{M} \models \neg\phi(\bar{d})$; thus, by ii), $\mathcal{N} \models \neg\phi(\bar{d})$, a contradiction. $\square$

Lemma 7.12. *Let T be an L-theory. Suppose that for every quantifier-free L-formula $\phi(\bar{v}, w)$ there is a quantifier-free formula $\psi(\bar{v})$ such that $T \models \exists w \phi(\bar{v}, w) \leftrightarrow \psi(\bar{v})$. Then, T has quantifier elimination.*

Proof. Let $\phi(\bar{v})$ be an L-formula. We wish to show that $T \models \forall \bar{v}(\phi(\bar{v}) \leftrightarrow \psi(\bar{v}))$ for some quantifier-free formula $\phi(\bar{v})$. We prove this by induction on the complexity of $\phi(\bar{v})$.

If ϕ is quantifier-free, there is nothing to prove. Suppose that for $i = 0, 1$, $T \models \forall \bar{v}(\theta_i(\bar{v} \leftrightarrow \psi_i(\bar{v})))$, where ψ_i is quantifier-free.

If $\phi(\bar{v}) = \neg\theta_0(\bar{v})$, then $T \models \forall \bar{v}(\phi(\bar{v}) \leftrightarrow \neg\psi_0(\bar{v}))$.

If $\phi(\bar{v}) = \theta_0(\bar{v}) \wedge \theta_1(\bar{v})$, then $T \models \forall v(\phi(\bar{v}) \leftrightarrow (\psi_0(\bar{v}) \wedge \psi_1(\bar{v})))$.

In either case, ϕ is equivalent to a quantifier-free formula.

Suppose that $T \models \forall \bar{v}(\theta(\bar{v}, w) \leftrightarrow \psi_0(\bar{v}, w))$, where ψ_0 is quantifier-free and $\phi(\bar{v}) = \exists w \theta(v,\bar{} w)$. Then $T \models \forall(\phi(\bar{v}) \leftrightarrow \exists w(\psi_0(\bar{v}, w)))$. By our assumptions, there is a quantifier-free $\psi(\bar{v})$ such that $T \models \forall \bar{v}(\exists w \psi_0(\bar{v}, w) \leftrightarrow \psi(\bar{v}))$. But then $T \models \forall \bar{v}(\phi(\bar{v} \leftrightarrow \psi(\bar{v})))$. $\square$

Corollary 7.13. *Let T be an L-theory. Suppose that for all quantifier-free formulas $\phi(\bar{v}, w)$, if $\mathcal{M}, \mathcal{N} \models T$, $\mathcal{A}$ is a common substructure of $\mathcal{M}$ and $\mathcal{N}$, $\bar{a} \in A$, and there is $b \in M$ such that $\mathcal{M} \models \phi(\bar{a}, b)$, then there is $c \in N$ such that $\mathcal{N} \models \phi(\bar{a}, c)$. Then, T has quantifier elimination.*

Theorem 7.14. *Dense linear ordering has quantifier elimination.*

Proof. Let A be a finite common substructure of two models O_1 and O_2. We choose an ascending enumeration $A = \{a_1, \ldots, a_n\}$. Let $\exists y \rho(y)$ be a simple existential $L(A) -$ *sentence*, which is true in O_1 and assume $O_1 \models \rho(b_1)$. We want to extend the order preserving map $a_i \mapsto a_i$ to an order preserving map $A \cup \{b_1\} \to O_2$. For this we have to find an image b_2 of b_1. There are four cases:

i) $b_1 \in A$. We set $b_2 = b_1$.

ii) b_1 lies between a_i and a_{i+1}. We choose b_2 in O_2 between a_i and a_{i+1}.

iii) b_1 is smaller than all elements of A. We choose b_2 also smaller than all elements of A.

iv) b_1 is bigger than all elements of A. We choose b_2 bigger than all elements of A.

This defines an isomorphism $A \cup \{b_1\} \to A \cup \{b_2\}$, which shows that $O_2 \models \rho(b_2)$. $\square$

Theorem 7.15. *(Tarski) The theory of algebraically closed fields has quantifier elimination.*

Definition 7.16. A field is *formally real* if -1 is not a sum of squares.

Definition 7.17. A linear ordering $<$ on R is compatible with the ring structure if for all $x, y, z \in R$

$$x < y \to x + z < y + z$$
$$x < y \wedge 0 < z \to xz < yz$$

A field $(K, <)$ together with a compatible ordering is an *ordered field.*

Lemma 7.18. *A field has a compatible ordering if and only if it is formally real.*

Proof. Suppose that K is an ordered field. It is easy to see that in an ordered field sums of squares can never be negative, hence formally real.

For the converse first notice that $\Sigma\Box$, the set of all sums of squares in K, is a *semi-positive cone*, i.e., a set P such that

$$\Sigma\Box \subset P$$
$$P + P \subset P$$
$$P \cdot P \subset P$$
$$-1 \notin P.$$

The first and third condition easily imply

$$x \in P \setminus 0 \implies \frac{1}{x} \in P.$$

Therefore, condition 4 is equivalent to $P \cup -P = 0$. It is also easy to see that for all b, the set $P + bP$ has the first three properties of a semi-positive cone. Condition 4 holds if and only if $b = 0$ or $-b \notin P$. We now choose P as a maximal semi-positive cone. Then

$$x \leq y \iff y - x \in P$$

and we obtain a compatible ordering of K. $\square$

Corollary 7.19. *Let K be a field not of characteristic 2 and let $a \in K$. There is an ordering of K making a negative if and only if a is not the sum of squares.*

Proof. If $a \notin \Sigma\Box$, then $\Sigma\Box - a \cdot \Sigma\Box$ is a semi-positive cone. $\Box$

Definition 7.20. An *ordered field* $(R, <)$ is real closed if

a) every positive element is a square.

b) every polynomial of odd degree has a zero.

We call $(R, <)$ a *real closure* of the subfield $(K, <)$ if R is real closed and algebraic over K.

Theorem 7.21. *Every ordered field $(K, <)$ has a real closure, and this is uniquely determined up to isomorphism over K.*

Theorem 7.22. *Let R be real closed. Then $C = R(\sqrt{-1})$ is algebraically closed.*

Theorem 7.23 (Hilbert's 17th Problem)**.** *Let $(K, <)$ be a real closed field. A polynomial $f \in K[X_1, \ldots, X_n]$ is a sum of squares*

$$f = g_1^2 + \ldots + g_k^2$$

of rational functions $g_i \in K(X_1, \ldots X_n)$ if and only if

$$f(a_1, \ldots, a_n) \geq 0$$

for all $a_1, \ldots, a_n \in K$.

1 Main Result

In the previous week we defined what it means for a group and a ring to be finitely axiomatizable(**FA**) and what it means for G to be interpreted in R (i.e. if there are two logical sentences α_1, α_2 s.t. $\alpha_1(R^{(d)})$ is isomorphic to G with the operation defined by α_2) we will say that this interpretation is strongly topological if the operation defined by α_2 is continuous in every profinite ring.
Similarly we have defined when R can be interpreted in G. With τ being the inclusion into $G^{(d)}$ and μ describing the operation. We can define a sentence ρ s.t. $G \vDash \rho(\bar{g})$ and for any group H with $H \vDash \rho(\bar{h})$ then $\tau(\bar{h}; H) =: S$ is a ring with the multiplication defined by μ.
We will call the interpretation strongly topological if S is a profinite ring.
Now for such H and $\bar{h}$, we have for each formula ϕ in L_{rg} some ϕ^* in L_{gp} s.t.

$$\tau(\bar{h}; H) \vDash \phi \iff H \vDash \phi^*(\bar{h})$$

If we have both of these and the functions:

$$\left\{ \theta : G = \hat{\alpha}(R) \to \hat{\alpha}(\hat{\tau}(R)) \subseteq G^{(d)} \theta' : R = \hat{\tau}(G) \to \hat{\tau}(\hat{\alpha}(G)) \subseteq R^{(d)} \right.$$

are definable.
We will call a group G **algebraically rigid** if each profinite group that is abstractly isomorphic to G is also topologically isomorphic (note that if G is FA then it is also **algebraically rigid**).
In this seminar we will show the following result:

Theorem 1. *For any complete, unramified regular local ring R with finite residue field, the profinite group $Af_1(R)$ is FA.*

The plan is to show that the rings in the theorem are FA and that for FA rings some bi-interpretable groups are also FA.

2 The Lemmas

Lemma 1. *Let R be an FA profinite ring. If R is strongly topologically interpreted in some G. Then there is some formula ψ s.t.*

- $G \vDash \psi(\bar{g})$ *(with $\bar{g}$ comming from the interpretation)*
- *for any profinite group H and tuple $\bar{h}$ s.t. $H \vDash \psi(\bar{h})$, then $\hat{\tau}(\bar{h}, H)$ is topologically isomorphic*

Proof. Take σ_R to be the defining sentence for R then we can set: $\psi(\bar{y}) = \rho(\bar{y}) \wedge \sigma_R^*(\bar{y})$ ☐

Lemma 2. *Take G, R a profinite group and ring that are bi-interpretable, with G being algebraically rigid. Then if R if FA and the interpretation of R in G is strongly topological, then G is also FA (in profinite groups).*

Proof. Take ψ as in the above lemma. Since the isomorphism θ from G to $\alpha(\tau(G))$ is definable we can express the fact that it is an ismorphism by some formula $\Theta(\bar{g})$. Set:

$$\Sigma_G(\bar{y}) \equiv \psi(\bar{y}) \wedge \Theta(\bar{y})$$

Clearly $G \vDash \Sigma_G(\bar{g})$.
Further take some group H s.t. $H \vDash \Sigma_G(\bar{h})$ for some $\bar{h}$. Since $H \vDash \psi(\bar{h})$ we have $S \coloneqq \hat{\tau}(H;\bar{h}) \cong R$. Thus $\hat{\alpha}(S)$ forms a group isomorphic to $\hat{\alpha}(R)$.
Since $H \vDash \Theta(\bar{h})$ the function defined by the same formula as θ is an isomorphism on H i.e. we have an isomorphism:

$$\tilde{\theta} : H \to \hat{\alpha}(S) \cong \hat{\alpha}(R) \cong \hat{\alpha}(\hat{\tau}(G))$$

so $\tilde{\theta}^{-1} \circ \theta$ is an isomorphism from G to H. Since G is algebraically rigid we also have a topological isomorphism. □

Now we need to find a class of profinite rings that are FA so that then we can show that some matrix groups on these rings are also FA.
Luckily we have a classification result for the class of regular, unramified, complete local rings with finite residue field. For this we define:

$$\mathfrak{o}_q = \mathbb{Z}_p[\zeta_{q-1}]$$

where $q = p^f$ for some prime p and ζ_{q-1} is a regular $(q-1)$st root of unity.

Lemma 3. *For any regular, unramified, complete local ring R with finite residue field $\mathbb{F}_q$ and dimension $d \geq 1$ we have one of the following:*

- $R \cong \mathbb{F}_q[[t_1, \dots, t_d]]$
- $R \cong \mathfrak{o}_q[[t_1, \dots, t_d]]$

This allows us to show that:

Lemma 4. *Any regular, unramified, complete local ring with finite residue field is FA*

With some extra ring theory magic we have:

Lemma 5. *For R a complete local ring with finite residue field, any finite index ideal in R is open.*

This allows us to conclude that such rings are algebraically rigid and even that $\mathrm{Af}_1(R)$ is algebraically rigid.

3 Proof

To finalize the proof of Theorem 1 we just have to show:

Theorem 2. *For any R that is FA the group $G = Af_1(R)$ is also FA.*

Proof. We can do this by showing that G and R are bi-intepretable with the interpretation of R in G being strongly topological.
Since:

$$G = \begin{pmatrix} 1 & R \\ 0 & R^* \end{pmatrix} \leq GL_2(R)$$

it is clear that G is interpretable in R.
To interpret R in G, we can define the following elements for each $\lambda \in R$ and $\xi \in R^*$:

$$u(\lambda) = \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix},\ h(\xi) = \begin{pmatrix} 1 & 0 \\ 0 & \xi \end{pmatrix}$$

Now we can define $u := u(1)$ and $h := h(r)$ for some $r \neq 1 \in R^*$. We have:

$$U := u(R) = C_G(u) = \mathrm{com}(u; G);$$
$$H := h(R^*) = C_G(h) = \mathrm{com}(h; G);$$

where com denotes the formula saying when two elements commute.
Furthermore we note that:

$$u(\lambda)^{h(\xi)} = u(\xi\lambda)$$

Th construct an interpretation of R in G we use the map $u : R \to U$. This is an isomorphism from $(R, +)$ to U. Now if we define the multiplication operation on U by:

$$u(\alpha) \cdot u(\beta) = u(\alpha\beta)$$

Now we have to define a formula μ s.t. for each $x, y, z \in U$:

$$x \cdot y = z \iff G \vDash \mu(u, h; x, y, z)$$

For this we note that for any $v = u(\beta) \in U$, if $\beta \in R^*$, we have $v = u^{h(\beta)}$ and if $\beta \in \mathfrak{m}$, we have $\beta + 1 \in R^*$ and thus:

$$[u, h(\beta + 1)] = u^{h(\beta+1)}u(-1) = u(\beta + 1)u(-1) = u(\beta) = V$$

So if we define μ as follows:

$$\begin{aligned} \mu(u, h, v_1, v_2, v_3) \equiv \exists x, y \in H(\mathrm{com}(x, h) \wedge \mathrm{com}(x, h)) \wedge \\ \Big((v_1 = u^x \wedge v_2 = u^y \wedge v_3 = u^{x,y}) \vee \\ (v_1 = u^x \wedge v_2 = [u, y] \wedge v_3 = [u^x, y]) \vee \\ (v_1 = [u, x] \wedge v_2 = u^y \wedge v_3 = [u^y, x]) \vee \\ (v_1 = [u, x] \wedge v_2 = [u, y] \wedge v_3 = [[u, y], x])\Big) \end{aligned}$$

To see that this interpretation is strongly topological we look at the sentence ρ (with two free parameters) that guarantees that μ defines a ring multiplication. Now let $\tilde{G}$ be a group that has $\tilde{u}, \tilde{v}$ s.t. $\tilde{G} \vDash \rho(\tilde{u}, \tilde{v})$. Now if we define $\tilde{U} = \mathrm{com}(\tilde{u}, \tilde{G})$ and $\tilde{H} = \mathrm{com}(\tilde{h}, \tilde{G})$. we first note that for any $x, y \in \tilde{H}$: $[x, y] \in \tilde{U}$.

So for any open normal subgroup N of $\tilde{G}$. Then for any $x, y, x', y' \in \tilde{H}$ if we have $u^x \equiv u^{x'} (\mod N)$ and $u^y \equiv u^{y'} (\mod N)$ we have:

$$u^{xy} \equiv u^{x'y} = u^{yx'} \equiv u^{y'x'} = u^{x'y'} \quad (\mod N)$$

We can do the same with the other parts of the definition of the multiplication. This shows us that if $v_1 \equiv v_1' (\mod N)$ and $v_2 \equiv v_2' (\mod N)$ we must have $v_1 \cdot v_2 \equiv v_1' \cdot v_2 (\mod N)$.
Thus by the profinitness of $\tilde{H}$ the operation $\cdot$ is continous.
Now we note that the functions θ, θ' look as follows:

$$\theta : \begin{pmatrix} 1 & \lambda \\ 0 & \xi \end{pmatrix} \mapsto (1, \lambda, 0, \xi) \mapsto (u(1), u(\lambda), u(0), u(\xi))$$

and:

$$\theta' : \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \mapsto (1, \lambda, 0, 1)$$

We can quickly see that θ' is definable. To see that θ is as well we define $\tilde{u}(g) = u(\lambda)$ and $\tilde{h}(g) = h(\xi)$. Since $g = \tilde{h}(g)\tilde{u}(g)$ and H, U have trivial intersection we have:

$$\{\tilde{u}(g)\} = Hg \cap U$$
$$\{\tilde{h}(g)\} = gU \cap H$$

this shows that $\tilde{u}$ and $\tilde{h}$ are definable functions(as U, H are) and since:

$$\theta(g) = (u, \tilde{u}(g), 1_G, u^{\tilde{h}(g)})$$

theta is also definable $\square$

O-MINIMAL STRUCTURES AND PILA-WILKIE COUNTING THEOREM

Zhenlin Ran

1 Week 9: O-minimal structures and some important theorems

We start by a thoerem of Tarski, which gives some heuristics of the concept of o-minimal structures.

Theorem 1.1. (Tarski) The theory $RCF_<$ of real closed fields in the language $\{+, -, 0, 1, <\}$ has quantifier elimination, i.e. given a formula $\phi(x_1, ..., x_n)$ there is a formula $\psi(x_1, ..., x_n)$ without quantifiers that is equivalent to $\phi(x_1, ..., x_n)$ in the model $RCF_<$. That means

$$R \models \forall x_1 \cdots x_n(\phi(x_1, ..., x_n) \leftrightarrow \psi(x_1, ..., x_n)).$$

Corollary 1.2. If $R_1 \subset R_2$ are real closed fields, then $R_1 \preccurlyeq R_2$ ("$\preccurlyeq$" means R_1 is an elementary structure of R_2, i.e. for every formula $\varphi(x_1, ..., x_n)$, R_2 satisfies φ whenever R_1 does. Or equivalently, if $a_1, ..., a_n \in R_1$, $R_1 \models \varphi(a_1, ..., a_n) \leftrightarrow R_2 \models \varphi(a_1, ..., a_n)$).

Corollary 1.3. If R is a real closed field, then every definable subset of R is a finite union of points and non-empty open intervals with extremities in $R \cup \{\pm\infty\}$.

O-minimal Structures

Definition 1.4. Let $\mathcal{L} = \{<, \cdots\}$ be a language containing a binary relation $<$, and $\mathcal{R}$ be a model of the $\mathcal{L}$-theory of dense linear order without endpoints. We say $\mathcal{R}$ is an *o-minimal structure* if every definable subset (with parameters in $\mathcal{R}$) of $\mathcal{R}$ is a union of finitely many points and finitely many open intervals.

Remark 1.5. The theory of dense linear order without endpoints can be defined by following sentences:

- $\forall x, y(x < y \vee x = y \vee y < x)$;
- $\forall x, y, z((x < y \wedge y < z) \rightarrow (x < z))$;
- $\forall x, y(x < y \rightarrow \exists z(x < z \wedge z < y))$;
- $\forall x(\exists y, z(y < x \wedge x < z))$.

We note that the dense linear order canonically generates a topology on $\mathcal{R}$. We fix the notations that $\mathrm{Cl}(\cdot)$ denotes the closure and $\mathrm{Int}(\cdot)$ denotes the interior. A bold letter means an n-ary. If $\mathcal{R}$ has domain R, we denote R_∞ by $R \cup \{-\infty, +\infty\}$.

Lemma 1.6. Let $\mathcal{R}$ be an o-minmal structure with domain R. Let $A \subset R$ and $B \subset R^n$ be definable subsets.

(1) $\inf(A)$ and $\sup(A)$ exist in R_∞ (Dedekind completeness for definable sets).

(2) The boundary $\mathrm{Bd}(A) : \{x \in R : \text{each interval containing } x \text{ intersects both } A \text{ and } R \backslash A\}$ is finite. If $a_1 < \cdots < a_k$ are the points of $\mathrm{Bd}(A)$ in order, then each interval (a_i, a_{i+1}), where $a_0 = -\infty$ and $a_{k+1} = +\infty$, is either part of A or disjoint from A.

(3) If A is infinite, it contains a non-empty interval.

(4) $\mathrm{Cl}(B)$ and $\mathrm{Int}(B)$ are both definable.

(5) If $B \subset C \subset R^n$ where C is definable and B is open in C, then there is an open definable U such that $C \cap U = B$.

Proof. $(1) - (3)$ are just immediate consequences of the definition of o-minimal structures. For (4), we note the following equivalence:

$$(x_1, ..., x_n) \in \mathrm{Cl}(B)$$
$$\Leftrightarrow$$
$$\forall y_1 \cdots \forall y_n \forall z_1 \cdots \forall z_n [(y_1 < x_1 < z_1 \wedge \cdots \wedge y_n < x_n < z_n) \rightarrow$$
$$\exists a_1 \cdots \exists a_n (y_1 < a_1 < z_1 \wedge \cdots y_n < a_n < z_n \wedge (a_1, ..., a_n) \in B)]$$

Now use the interpretations of logical symbols in terms of operations on sets. The proof to $\mathrm{Int}(B)$ is similar. For (5), note that one can take U for the union of all boxes in R^n whose intersection with C is contained in B. □

Definition 1.7. Let $\mathcal{R}$ be an o-minimal structure with domain R. A set $X \subset R^n$ is called *definably connected* if X is definable and X is not the union of two disjoint nonempty definable open subsets of X. A function $f : X \to Y$ is said to be *definable* if its graph $\Gamma(f)$ is definable.

Lemma 1.8. (1) The definably connected subsets of R are the following: the empty set, the intervals, the sets $[a, b)$ with $-\infty < a < b \leqslant +\infty$, the sets $(a, b]$ with $-\infty \leqslant a < b < +\infty$, and the sets $[a, b]$ with $-\infty < a \leqslant b < +\infty$.

(2) The image of a definably connected set $X \subset R^n$ under a definable continuous map $f : X \to R^m$ is definably connected.

(3) If X and Y are definable subsets of R^n, $X \subset Y \subset \mathrm{Cl}(X)$, and X is definably connected, then Y is definably connected.

(4) If X and Y are definably connected subsets of R^n and $X \cap Y \neq \varnothing$, then $X \cup Y$ is definably connected.

Proof. (1) is trivial. To see (2), we first note that $\Gamma(f) \subset R^n \times R^m$ is definable, which implies that $f(X)$ is definable. To see it's definably connected, we prove that for any definable subset $Y \subset R^m$, $f^{-1}(Y)$ is definable. Indeed, $R^n \times Y$ is definable, hence so is

$$\Gamma(f) \cap (R^n \times Y) = \{(x, y) \in R^n \times R^m : x \in X, f(x) = y \in Y\}.$$

Thus, $f^{-1}(Y)$ is definable, which completes our proof to the second statement.

To prove (3), we assume $Y = Y_1 \cup Y_2$ to be the disjoint union of two non-empty definable open subsets Y_1 and Y_2. We note that the inclusion map is definably continuous. Therefore, by the proof of (2), either Y_1 or Y_2 contains X. This means that one of Y_1 and Y_2 is dense in Y as $Y \subset \mathrm{Cl}(X)$, which implies that $Y_1 \cap Y_2 \neq \varnothing$. Contradiction.

Now we prove (4). Assume $X \cup Y = W_1 \cup W_2$, where W_1 and W_2 are two disjoint non-empty definable open subsets. As $X \cap (W_1 \cup W_2) = X$, we see either W_1 or W_2 contains X. The same is true for Y. If X and Y lie in the same open subset, then we are done. Otherwise, suppose $X \subset W_1$ and $Y \subset W_2$. Since $W_1 \cap W_2 = \varnothing$, we have $X \cap Y = \varnothing$, which is a contradiction. $\square$

We will use definable Ramsey theorem to prove the monotonicity theorem, which will make our proof much shorter. However, the proof to the definable Ramsey theorem is omitted. From now on, we always assume R is the domain of some o-minimal structure unless specified otherwise.

Theorem 1.9. (*Definable Ramsey Theorem*) Let $I \subset R$ be an open interval, and assume $I^2 = I \times I = X_1 \cup \cdots \cup X_r$ with each X_i definable. Then there is a non-empty open interval $J \subset I$ and $i \in \{1, ..., r\}$ such that if $a < b \in J$, then $(a, b) \in X_i$.

Theorem 1.10. (*Monotonicity Theorem*) Let $f : I \to R$ be a definable function on the interval $I := (a.b)$. Then there are points $a = a_0 < a_1 < \cdots < a_k < a_{k+1} = b$ such that on each (a_i, a_{i+1}) the function f is continuous, and either constant or strictly monotone.

Proof. Let

$$\begin{aligned} A_1 &= \{x \in I : f \text{ is locally constant at } x\}; \\ A_2 &= \{x \in I : f \text{ is locally strictly increasing at } x\}; \\ A_3 &= \{x \in I : f \text{ is locally strictly decreasing at } x\}. \end{aligned}$$

It is not hard check that A_i for any $i \in \{1, 2, 3\}$ is definably open. The openness is trivial. To see they are definable, we show only for A_1 for example. Indeed, A_1 can be defined by the formula

$$\exists y_1 \exists y_2((a < y_1 < x < y_2 < b) \wedge (\forall z(y_1 < z < y_2) \to f(z) = f(x))).$$

Claim: $B := I \backslash (A_1 \cup A_2 \cup A_3)$ is finite.

If not, it then contains an open interval J as B is definable. We set

$$\begin{aligned} X_1 &= \{(x, y) \in J^2 : f(x) = f(y)\}; \\ X_2 &= \{(x, y) \in J^2 : f(x) < f(y)\}; \\ X_3 &= \{(x, y) \in J^2 : f(x) > f(y)\}. \end{aligned}$$

They form a partition of J^2. By Theorem 1.9, there is an open interval $J_0 \subset J$ and $i \in \{1, 2, 3\}$ such that if $c < d$ are in J_0 then $(c, d) \in X_i$. So $J_0 \subset A_i$. This is a contradciton. So B is finite.

Since all A_i's are open and B is finite, we are only left to show that f is continuous on small intervals. Thus, we may assume $f : I \to R$ is constant or strictly monotone. If f is constant, it is trivial. Otherwise $f(I)$ is infinite. Let $J \subset f(I)$ be an open interval. It is not hard to check by the monotonicity that $f^{-1}(J)$ is an open interval. By the o-minimality we see f is continuous on I, possibly excluding some finite set. We complete our poof by adding these finitely many points to B and then ordering them to get $a_1 < \cdots < a_k$. □

Corollary 1.11. Let $f : (a, b) \to R$ be definable. Then for each $c \in (a, b)$ the limits $\lim_{x \to c^-} f(x)$ and $\lim_{x \to c^+} f(x)$ exist in R_∞. Also the limits $\lim_{x \to a^+} f(x)$ and $\lim_{x \to b^-} f(x)$ exist in R_∞.

Corollary 1.12. Let $f : [a, b] \to R$ be continuous and definable. Then f takes a maximum and a minimum value in $[a, b]$.

Let $A \subset R^2$. We denote by $A_x = \{b \in R : (x, b) \in A\}$ the fibre of $x \in A$.

Definition 1.13. We say $A \subset R^2$ is *scattered* if $\{x \in R : A_x \text{ is infinite}\}$ is finite.

Theorem 1.14. (Finiteness Lemma) If $A \subset R^2$ is definable, then there is some $N \in \mathbb{N}$ such that for any $x \in R$, A_x is either infinite or finite of size $\leqslant N$.

Proof. The proof is rather long. We omit it. □

Theorem 1.15. If $A \subset R^2$ is definable and scattered, then there is a bound N on the size of finite A_x.

Remark 1.16. Indeed, Theorem 1.14 and Theorem 1.15 are equivalent. It is trivial that Theorem 1.14 implies Theorem 1.15. To show the other way around, we first note that $\{x \in R : A_x \text{ is infinite}\}$ is definable. This is because if A is defined by $\phi(x, y)$ then "A_x is infinite" is equivalent to

$$\exists y_1 \exists y_2 \forall y (y_1 < y < y_2 \to \phi(x, y)).$$

Then $\pi_1(A) = \{x \in R : A_x \text{ is infinite}\} \cup \{x \in R : A_x \text{ is finite}\}$. The first subset if definable, hence a finite union of open intervals and points. We can then get $A' \subset A$ which is scattered and definable by removing the open intervals appeared in the first subset of $\pi_1(A)$. Now the equivalence of two theorems is obvious.

Cell Decomposition

Definition 1.17. We define by induction the notion of a *cell* $C \subset R^n$ and of associated sequence $(i_1, ..., i_n)$ of 0 and 1 as follows:

- When $n = 1$:
  (1) a singleton $\{a\}$ is a cell. In this case $i_1 = 0$;
  (2) a non-empty open interval (a, b) is a cell for $a, b \in R_\infty$. In this case $i_1 = 1$.
- Assume we have defined cells in R^{n-1}. A subset $C \subset R^n$ is a cell if
  (1) The projection $\pi(C)$ is a cell in R^{n-1}, where $\pi : R^n \to R^{n-1}$ is given by $(x_1, ..., x_n) \mapsto (x_1, ..., x_{n-1})$.
  (2) Either $C = \Gamma(f)$, the graph of $f : \pi(C) \to R$ which is continuous and definable. In this case $i_n = 0$;

(3) Or $C = (f, g)$, where $f, g : \pi(C) \to R \cup \{\pm\infty\}$ are continuous and definable on $\pi(C)$ such that $f(x) < g(x)$ whenever $x \in \pi(C)$, and

$$(f, g) := \{(x, y) \in R^n : x \in \pi(C), f(x) < y < g(x)\}.$$

In this case, $i_n = 1$.

By convention, R^0 is a cell. If C is a cell in R^n, we define the *dimension* of C to be $\dim(C) := \sum_{j=1}^{n} i_j$, where $(i_1, ..., i_n)$ is the associated sequence.

Remark 1.18. (1) A cell $C \subset R^n$ is open if and only if $\dim(C) = n$.

(2) Let C be a cell. If $\dim(C) < n$, then $\mathrm{Int}(C) = \varnothing$.

(3) If π_m is the projection on the first m coordinates $(m < n)$, and $C \subset R^n$ is a cell, then $\pi_m(C)$ is a cell. And if $a \in \pi_m(C)$, then C_a is a cell.

(4) A finite union of cells in R^n of dimension $< n$ has empty interior.

(5) If C is a cell, then it is definably connected.

(6) A cell is locally closed, i.e. open in its closure.

Definition 1.19. A *decomposition* $\mathfrak{D}$ of R^n is a partition of R^n defined by induction:

(1) A decomposition of R is a collection

$$\{(-\infty, a_1), (a_1, a_2), ..., (a_k, +\infty), \{a_1\}, ..., \{a_k\}\},$$

where $a_1 < \cdots a_k$ are points in R.

(2) A decomposition $\mathfrak{D}$ of R^n is a partition of R^n into finitely many cells $C_1, ..., C_k$ such that if $\pi : R^n \to R^{n-1}$ is the projection on the first $n - 1$ coordinates, then $\{\pi(C_1), ..., \pi(C_k)\}$ is a decomposition of R^{n-1}.

Let $S \subset R^n$ be a definable subset, and $\mathfrak{D} = \{C_1, ..., C_k\}$ be a decomposition of R^n. We say $\mathfrak{D}$ *partitions* S if $C_i \subset S$ or $C_i \cap S = \varnothing$ for any $i \in \{1, .., k\}$

Theorem 1.20. (Cell Decomposition Theorem)

(1) If $A_1, ..., A_k \subset R^n$ are definable, then there is a decomposition $\mathfrak{D}$ of R^n that partitions each A_i.

(2) If $f : A \to R$ is a definable function where $A \subset R^n$ is definable, then there is a decomposition $\mathfrak{D}$ of R^n which partitions A and such that if $B \in \mathfrak{D}$ and $B \subset A$, then $f|_B$ is continuous.

(3) (Uniform Finiteness Property) If $A \subset R^n$ is definable, and A_x is finite for every $x \in R^n$, then there is $N \in \mathbb{N}$ such that $|A_x| \leqslant N$ for all $x \in R^{n-1}$.

Remark 1.21. When $n = 1$ in (1), we can construct such $\mathfrak{D}$ by o-minimality. When $n = 1$ in (2), we can easily construct $\mathfrak{D}$ as well using the monotonicity thoerem. When $n = 2$, (3) is just Theorem 1.14

Theorem 1.22. If $X \subset R^n$ is a non-empty definable subset of R^n, then X has only finitely many definably connected components. They are open and closed in X, and they partition X.

2 Week 10: Definable invariants and analytic structures

As before, we fix an o-minimal structure $\mathcal{R}$ with domain R.

Dimensions

Definition 2.1. Let $X \subset R^n$ be a non-empty definable set. The *dimension of* X is

$$\dim X := \max\{i_1 + \cdots + i_n : X \text{ contains an } (i_1, ..., i_n)\text{-cell}\}.$$

To the empty set we assign the dimension $-\infty$.

Remark 2.2. We can also take a cell decomposition $\mathfrak{D}$ of X and take the maximal dimensions of the cells of $\mathfrak{D}$ contained in X to be the dimension of X. Also, $\dim X$ is the largest m such that for some projections $\pi : R^n \to R^m$ has non-empty interior.

Lemma 2.3. If $A \subset R^n$ is an open cell and $f : A \to R^n$ an injective definable map, then $f(A)$ contains an open cell

Proof. By induction. Left as exercise for interested readers. $\square$

Define $p_i : R^n \to R^k$ as follows: let $\lambda(1) < \cdots < \lambda(k)$ be the indices $\lambda \in \{1, ..., n\}$ for which $i_\lambda = 1$, so that $k = i_1 + \cdots + i_n$; then

$$p_i(x_1, ..., x_n) := (x_{\lambda(1)}, ..., x_{\lambda(k)}).$$

It is easy to show by induction on n that p_i maps each i-cell A homeomorphically onto an open cell $p_i(A)$ in R^k. We denote $p_i(A)$ also by $p(A)$ and the homeomorphism $p_i|_A : A \to p(A)$. Clearly $p_A = id_A$ if A is an open cell.

Proposition 2.4. (1) If $X \subset Y \subset R^n$ are definable, then $\dim X \leqslant \dim Y \leqslant n$.

(2) If $X \subset R^m$ and $Y \subset R^n$ are definable and there is a definable bijection between X and Y, then $\dim X = \dim Y$.

(3) If $X, Y \subset R^n$ are definable, then $\dim(X \cup Y) = \max\{\dim X, \dim Y\}$.

Proof. (1) is obvious. To prove (2), let $f : X \to Y$ be a definable bijection and $d = \dim X$, $e = \dim Y$. It suffices to show $d \leqslant e$. Let A be an $(i_1, ..., i_m)$-cell contained in X, with

$$d = i_1 + \cdots + i_m.$$

Then $f \circ (p_A^{-1}) : p(A) \to Y$ is an injective map and $p(A)$ an open cell. Replacing X by $p(A)$, Y by $f(A)$ and f by $f \circ (p_A^{-1})$ we may as well assume that $d = m$ and that X is an open cell in R^d. Let

$$Y = C_1 \cup \cdots \cup C_k$$

be a partition of $Y = f(X)$ into cells. Then

$$X = f^{-1}(C_1) \cup \cdots \cup f^{-1}(C_k).$$

We note $f^{-1}(C_i)$ contains an open cell B for some i. This is because there is a decomposition $\mathfrak{D}$ of R^d that partitions each $f^{-1}(C_i)$ by Theorem 1.20 and then use Remark 2.2. Fix such i and B. Let $C_i = C \subset R^n$ be a $(j_1, ..., j_n)$-cell. We shall prove that

$$d \leqslant j_1 + \cdots + j_n.$$

Assume not, i.e. $d > j_1 + \cdots + j_n$. The decomposition

$$B \xrightarrow{f|_B} C \xrightarrow{p_C} p(C) \subset R^{j_1+\cdots+j_n}$$

is an injective definable map. Indentifying $R^{j_1+\cdots+j_n}$ with a non-open cell $(R^{j_1+\cdots+j_n}) \times \{P\}$ in R^d, where $P \in R^{d-(j_1+\cdots j_n)}$, we obtain a contradiction with Lemma 2.3.

To prove (3), let $d = \dim(X \cup Y)$ an let A be an $(i_1, ..., i_n)$-cell contained in $X \cup Y$, with $d = i_1 + \cdots + i_m$. The open cell $p(A) \subset R^d$ is the union of $p_A(A \cap X)$ and $p_A(A \cap Y)$. By the cell decomposition theorem, one of those sets, say $p_A(A \cap X)$, contains a box B in R^d. Then $p_A^{-1}(B)$ is an $(i_1, ..., i_n)$-cell contained in X, so that

$$\dim X \geqslant d \geqslant \dim X \;\Rightarrow\; \dim X = \dim(X \cup Y).$$

Thus, we are done. $\square$

The following result is about the dimension of definable family.

Proposition 2.5. Let $S \subset R^m \times R^n$ be definable. For $d \in \{-\infty, 0, 1, .., n\}$ put

$$S(d) := \{a \in R^m : \dim S_a = d\}.$$

Then $S(d)$ is definable and the part of S above $S(d)$ has dimension given by

$$\dim\left(\bigcup_{a \in S(d)} \{a\} \times S_a\right) = \dim(S(d)) + d.$$

Proof. Omitted. $\square$

We further talk about an important theorem about the dimension of o-minimal ordered group. Let $\mathcal{L}$ be the language $\{+, -, 0, 1, <\}$ of ordered group. Let $\mathcal{R}$ be an $\mathcal{L}$-structure with domain R such that R is an ordered group which is also o-minimal. Note that every such group is divisible and abelian.

Theorem 2.6. If $A \subset R^n$ is definable, then $\dim(\mathrm{Cl}(A) \backslash A) < \dim(A)$.

Proof. Omitted. $\square$

Corollary 2.7. (1) Let $S \subset T \subset R^n$ be definable and $\dim(S) = \dim(T) \geqslant 0$. Then S has non-empty interior in T (for the induced topology), and

$$\dim(S \backslash \mathrm{Int}_T(S)) < \dim S.$$

(2) If $A \subset R^n$ is definable, then $\dim(\mathrm{Cl}(A) \backslash \mathrm{Int}(A)) < n$

Euler Characteristic

Euler characteristic is another subtle definable invariant. Observe that a partition of an interval into cells consists necessarily k points and $k+1$ intervals for some $k \geqslant 0$, and that the quantity $k-(k+1)=-1$ is independent of the partition considered.

For each cell C of dimension d, we assign

$$E(C) := (-1)^d.$$

If S is a definable set in R^n and $\mathfrak{D}$ is a finite partition of S into cells, we assign

$$E_{\mathfrak{D}}(S) = \sum_{C \in \mathfrak{D}} E(C) = k_0 - k_1 + \cdots + (-1)^d k_d + \cdots (-1)^n k_n,$$

where k_d is the number of cells of dimension d in $\mathfrak{D}$.

Definition 2.8. Let S be a definable subset of R^n. The *Euler characteristic* $E(S)$ is $E_{\mathfrak{D}}(S)$ for some finite partition $\mathfrak{D}$ of S into cells.

We need to show this definition is well-defined, i.e. $E_{\mathfrak{D}_1}(S) = E_{\mathfrak{D}_2}(S)$ for any two finite partitions $\mathfrak{D}_1$ and $\mathfrak{D}_2$.

Lemma 2.9. If $\mathfrak{D}$ is a finite partition of a cell C with dimension d, then

$$E_{\mathfrak{D}}(C) = E(C) = (-1)^d.$$

Proof. Suppose $C \subset R^n$. This can be done by induction on n. It is trivial when $n=1$. Let $C \subset R^{n+1}$ and assume inductively

$$E_{\pi(\mathfrak{D})}(\pi(C)) = E(\pi(C)). \tag{1}$$

We prove the case that C is an $(i_1, ..., i_n, 1)$-cell. The case that C is an $(i_1, ..., i_n, 0)$-cell is similar.

Let $B \in \pi(\mathfrak{D})$, so the list of distinct cells of $\mathfrak{D}$ that map onto B is

$$\Gamma(f_1), ..., \Gamma(f_t), (f_0, f_1), ..., (f_t, f_{t+1}),$$

for certain continuous definable functions f_i on B. Let $\dim B = d$. The contribution of this list of cells to $E_{\mathfrak{D}}(C)$ equals to:

$$t \cdot (-1)^d + (t+1) \cdot (-1)^{d+1} = (-1)^{d+1} = -E(B).$$

Summing over the various $B \in \pi(\mathfrak{D})$ and using (1) we get

$$E_{\mathfrak{D}}(C) = -\sum_{B \in \pi(\mathfrak{D})} E(B) = -E_{\pi(\mathfrak{D})}(\pi(C)) = -E(\pi(C)) = E(C).$$

The last equality comes from that C is an $(i_1, ..., i_n, 1)$-cell. $\square$

Lemma 2.10. Every two partitions $\mathfrak{D}_1$ and $\mathfrak{D}_2$ of a definable set $S \subset R^n$ into cells have a common refinement into a parititon $\mathfrak{D}$ of S into cells such that for each cell $C \in \mathfrak{D}_1 \cup \mathfrak{D}_2$ the restriction $\mathfrak{D}|C$ is a decomposition of C.

Proof. Application of cell decomposition theorem 1.20. □

Proposition 2.11. The defintion of Euler characteristic of a definable set $S \subset R^n$ is well-defined.

Proof. Let $\mathfrak{D}_1$ and $\mathfrak{D}_2$ be finite partitions of S into cells. We have to show

$$E_{\mathfrak{D}_1}(S) = E_{\mathfrak{D}_2}(S).$$

Take a common refinement $\mathfrak{D}$ of $\mathfrak{D}_1$ and $\mathfrak{D}_2$ using Lemma 2.10. Then

$$\begin{aligned} E_{\mathfrak{D}}(S) &= \sum_{C \in \mathfrak{D}_1} E_{\mathfrak{D}|C}(C) \\ &= \sum_{C \in \mathfrak{D}_1} E(C), \text{ by Lemma 2.9} \\ &= E_{\mathfrak{D}_1}(S). \end{aligned}$$

And using the same argument we get $E_{\mathfrak{D}}(S) = E_{\mathfrak{D}_2}(S)$. Thus we are done. □

Analytic structures

Let $\mathcal{R}$ be an o-minimal structure which is a model of real closed field. We denote by R the domain of R.

Definition 2.12. Let I be an open interval. A function $f : I \to R$ is *differentiable* at $x_0 \in I$ with *derivative* a if

$$\lim_{t \to 0} \frac{f(x_0 + t) - f(x_0)}{t} = a.$$

As in the case of real numbers, if f is differentiable at x_0 then it is continuous at x_0 and a is unique. We denote $a = f'(x_0)$.

Definition 2.13. Let $U \subset R^m$ be open, and $f = (f_1, ..., f_n) : U \to R^n$. Let $T : R^m \to R^n$ be a linear map. We say f is *differentiable* at $x_0 \in U$ with *differential* T if for any $\epsilon > 0$ there exists $\delta > 0$ such that if $\|x - x_0\| < \delta$ then

$$\|f(x) - f(x_0) - T(x - x_0)\| < \epsilon \|x - x_0\|.$$

We denote $T = D_{x_0}(f)$. If $f = (f_1, ..., f_n)$ is differentiable at x_0, then $\frac{\partial f_i}{\partial x_j}(x_0)$ exists for any i, j.. Actually,

$$T = \left(\frac{\partial f_i}{\partial x_j}(x_0) \right).$$

The classical results from calculus are still true. For example, Rolle's theorem, mean value theorem, inverse function theorem, implicit function theorem...

Theorem 2.14. Let $f : I \to R$ be a definable function and I is an interval. Then the set of points in I where f is not differentiable is finite.

Let $f : U \to R \subset \mathbb{R}$ be a function of C^{∞}-differentiable, where $U \subset \mathbb{R}^n$ is open. For $a \in U$, we have the *Taylor series*:

$$T_a(f) := \sum_{\alpha \in \mathbb{N}^n} \frac{1}{\alpha!} \cdot \frac{\partial^{\alpha} f}{\partial x^{\alpha}}(x-a)^{\alpha},$$

where $\alpha = (\alpha_1, ..., \alpha_n)$ and $x = (x_1, ..., x_n)$ such that

$$\alpha! = \alpha_1! \cdots \alpha_n!,\ \frac{\partial^{\alpha} f}{\partial x^{\alpha}} = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial x_2^{\alpha_2}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} f,\ (x-a)^{\alpha} = \prod (x_i - a_i)^{\alpha_i}.$$

Definition 2.15. We say f is *analytic at a* if the Taylor series $T_a(f)$ converges near a to $f(a)$, i.e. it converges on an open ball containing a. If f is analytic at every point of U, we say f is *analytic on U*.

Let $\mathbb{R}[[X_1, ..., X_m]]$ be the set of formal power series with variables $X_1, ..., X_m$ that converge on an open neighbourhood of $I^m := [-1, 1]^m$. If $f \in \mathbb{R}[[X_1, ..., X_m]]$, we set

$$\tilde{f}(x) = \begin{cases} f(x), & \text{if } x \in I^m. \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

We define the language $\mathcal{L}_{an} := \{+, -, \cdot, 0, 1, <, \tilde{f}\}_{m \in \mathbb{N},\ f \in \mathbb{R}[[X_1,...,X_m]]}$, and the *analytic structure*:

$$\mathbb{R}_{an} := (\mathbb{R}, +, -, \cdot, 0, 1, <, \tilde{f})_{m \in \mathbb{N},\ f \in \mathbb{R}[[X_1,...,X_m]]}.$$

Definition 2.16. (1) Let $X \subset \mathbb{R}^m$. We say X is *semi-analytic* at $x \in \mathbb{R}^m$ if x has an open neighborhood U s.t. $U \cap X$ is a finite union of sets of the form

$$\{y \in U : f(y) = 0, g_1(y) > 0, ..., g_k(y) > 0\},$$

where $f, g_1, ..., g_k$ are analytic on U.

(2) We say $X \subset \mathbb{R}^m$ is *semi-analytic* if it is smei-analytic at every point of $\mathbb{R}^m$.

(3) We say $X \subset \mathbb{R}^m$ is *subanalytic* at x if there is an open set U containing x and a bounded semi-analytic $S \subset \mathbb{R}^{m+n}$ s.t. $U \cap X = \pi(S)$, where $\pi : \mathbb{R}^{m+n} \to \mathbb{R}^m$ is the projection.

(4) We say X is *subanalytic* if it is subanalytic at every $x \in X$.

We collect some important properties:

(1) When $m = 1, 2$, then subanalytic implies semi-analytic. It is not true when $m > 2$.

(2) A bounded semi-analytic set has finitely many connected components.

(3) The bounded subanalytic subsets of $\mathbb{R}^m$ are of the form $\pi(S)$, where S is bounded semi-analytc of $\mathbb{R}^{m+n}$. A bounded subanalytic set therefore has only a finite number of connected components, and they are subanalytic.

(4) If $X \subset \mathbb{R}^m$ is subanalytic, then so is $\mathbb{R}^m \backslash X$.

(5) The closure of semi-analytic (resp. subanalytic) is also semi-analytic (resp. subanalytic).

Theorem 2.17. $\mathbb{R}_{an}$ is o-minimal.

Definition 2.18. Let $U \subset \mathbb{R}^m$ be an open subset. Let $(f_1, ..., f_r)$ be a sequence of real analytic functions on U, where $f_i : U \to \mathbb{R}$. We say $(f_1, ..., f_r)$ is a *Pfaffian chain of order* r if for all $1 \leqslant i \leqslant r$ and $1 \leqslant j \leqslant m$, then

$$\frac{\partial^j f_i}{\partial x_j}(\mathbf{x}) \in \mathbb{R}[x_1, ..., x_m, f_1(\mathbf{x}), ..., f_i(\mathbf{x})].$$

A function $f : U \to \mathbb{R}$ is *Pfaffian* if it is a part of a Pfaffian chain.

Set $\mathbb{R}_{exp} := \{\mathbb{R}, +, -, \cdot, 0, 1, <, \exp\}$, where exp is the exponential function.

Theorem 2.19. (Wilkie, 1996) $\mathbb{R}_{exp}$ is o-minimal and model complete.

Remark 2.20. By model complete we mean every substructure is an elementary substructure.

Conjecture 2.21. (Schanuel's conjecture) If $a_1, ..., a_n \in \mathbb{C}$ are $\mathbb{Q}$-linearly independent. Then

$$\text{tr.deg}(\mathbb{Q}(a_1, ..., a_n, \exp(a_1), ..., \exp(a_n))) \geqslant n.$$

3 Week 11: Pila-Wilkie Theorem

Notations

Throughout, $d, e, k, l, m, n \in \mathbb{N} = \{0, 1, 2, ...\}$, and $\epsilon, c, K \in \mathbb{R}^{>} := \{t \in \mathbb{R} : t > 0\}$. For $\alpha = (\alpha_1, ..., \alpha_m) \in \mathbb{N}^m$ we set

$$|\alpha| := \alpha_1 + \cdots + \alpha_m,$$

and given a field F (often $F = \mathbb{R}$) we set

$$x^\alpha := x_1^{\alpha_1} \cdots x_m^{\alpha_m}$$

for the usual coordinate functions $x_1, ..., x_m$ on F^m. Likewise, for any point $a = (a_1, ..., a_m) \in F^m$, we set

$$a^\alpha = a_1^{\alpha_1} \cdots a_m^{\alpha_m}.$$

Let $U \subset \mathbb{R}^m$ be open. For a function $f : U \to \mathbb{R}$ of class C^k, and for any $\alpha \in \mathbb{N}^m$ and $|\alpha| \leqslant k$ let

$$f^{(\alpha)} := \frac{\partial^{|\alpha|}}{\partial x^\alpha} f$$

denote the corresponding partial derivative of order α. We extend the above to C^k-maps $f = (f_1, ..., f_n) : U \to \mathbb{R}^n$ by

$$f^{(\alpha)} := (f_1^{(\alpha)}, ..., f_n^{(\alpha)}) : U \to \mathbb{R}^n$$

for α as before. This includes the case $m = 0$, where $\mathbb{R}^0$ has justone point and any map $f : U \to \mathbb{R}$ is of class C^k for all k, with $f^{(\alpha)} = f$ for the unique $\alpha \in \mathbb{N}^0$. For $a_1, ..., a_n \in \mathbb{R}^{\geqslant 0} := \{t \in \mathbb{R} : t \geqslant 0\}$ the number $\max\{a_1, ..., a_n\} \in \mathbb{R}^{\geqslant}$ equals 0 by convention if $n = 0$. For $(a_1, ..., a_n) \in \mathbb{R}^n$, we set

$$|a| := \max\{|a_1|, ..., |a_n|\}$$

in $\mathbb{R}^{\geqslant}$. This conflicts with our notation $|\alpha|$ for $\alpha \in \mathbb{N}^n$, but in practise no confusion will arise.

Pila-Wilkie theorem

Definition 3.1. We define the multiplicative height function $H : \mathbb{Q} \to \mathbb{R}$ by

$$H\left(\frac{a}{b}\right) := \max\{|a|, |b|\} \in \mathbb{N}^{\geqslant 1}$$

for coprime $a, b \in \mathbb{Z}$ and $b \neq 0$. For $a = (a_1, ..., a_n) \in \mathbb{Q}^n$,

$$H(a) := \max\{H(a_1), ..., H(a_n)\} \in \mathbb{N}.$$

Thus, $H(0) = H(1) = H(-1) = 1$, and for $q \in \mathbb{Q}$ we have $H(q) \geqslant 2$ if $q \notin \{0, 1, -1\}$, $H(q) = H(-q)$ and $H(q) = H(q^{-1})$ for $q \neq 0$.

Let $X \subset \mathbb{R}^n$. We set $X(\mathbb{Q}) := X \cap \mathbb{Q}^n$. Throughout T ranges over real numbers $\geqslant 1$, and we set

$$X(\mathbb{Q}, T) := \{x \in X(\mathbb{Q}) : H(a) \leqslant T\}$$

to be the (finite) set of rational points of X of height $\leqslant T$. Also, we set

$$N(X, T) := \#X(\mathbb{Q}, T) \in \mathbb{N}.$$

Definition 3.2. Let $X \subset \mathbb{R}^n$. The *algebraic part* of X, denoted by X^{alg}, is the union of the connected infinite semialgebraic subsets of X. The transcendental part of X is $X^{\mathrm{tr}} := X \backslash X^{\mathrm{alg}}$.

Remark 3.3. For $n \geqslant 1$, the interior of X (in $\mathbb{R}^n$) is part of X^{alg}.

Example 3.4. The set $X := \{(a, b, c) \in \mathbb{R}^3 : 1 < a, b < 2, c = a^b\}$ is definable in the o-minimal field $\mathbb{R}_{exp}$. For rational $q \in (1, 2)$, we have a semialgebraic curves

$$X_q := \{(a, q, c) : c = a^q\} \subset X,$$

and X^{alg} is the union of those X_q (not trivial).

We are ready to state the Pila-Wilkie theorem, also called the *Counting Theorem*:

Theorem 3.5. (Pila-Wilkie) Let $X \subset \mathbb{R}^n$ with $n \geqslant 1$ be definable in some o-minimal expansion of $\mathbb{R}$. Then for all ϵ there is a c such that for all T,

$$N(X^{\mathrm{tr}}, T) \leqslant cT^{\epsilon}.$$

Roughly speaking, it says there are few rational points on the transcendental part of a set definable in an o-minimal expansion of $\mathbb{R}$: the number of such points grows slower than any power T^{ϵ} with T bounding the height. To apply the counting theorem one needs to describe X^{alg} in some useful way. This typically involves Ax-Schanuel type transcendence results.

Ax-Schanuel type results

Though the Ax-Schanuel theorem is not the goal of this talk, we introduce it for the interested reader. We first recall a conjecture from Schanuel we mentioned in last week:

Conjecture 3.6. (Schanuel's conjecture) If $a_1, ..., a_n \in \mathbb{C}$ are $\mathbb{Q}$-linearly independent. Then

$$\mathrm{tr.deg}(\mathbb{Q}(a_1, ..., a_n, \exp(a_1), ..., \exp(a_n))) \geqslant n.$$

Remark 3.7. Actually suitable Grothendieck's conjecture implies Schanuel's conjecture.

Replace $\mathbb{Q}$ (resp. $\mathbb{C}$) by $\mathbb{C}$ (resp. $\mathbb{C}[[t_1, ..., t_m]]$), in 1971 Ax proved the following theorem:

Theorem 3.8. (Ax) Let $f_1, ..., f_n \in \mathbb{C}[[t_1, ..., t_m]]$ be power series over $\mathbb{C}$ with m variables which have no constant term and be linearly independent over $\mathbb{Q}$. Then

$$\mathrm{tr.deg}_{\mathbb{C}}\, \mathbb{C}(f_1, ..., f_n, \exp(f_1), ..., \exp(f_n)) \geqslant n + \mathrm{rank}(J),$$

where J is the Jacobian matrix: $J_{ij} = \frac{\partial f_i}{\partial t_j}$.

Geometric equivalent formulation: Let $W \subset \mathbb{C}^n \times (\mathbb{C}^*)^n$ be an irreducible algebraic subvariety. Let U be an irreducible analytic component of $W \cap \Pi$, where Π is the graph of the exponential map. Assume that the projection of U to $(\mathbb{C}^*)^n$ is not contained in a translate of any proper algebraic subgroup. Then $\dim W = \dim U + n$.

Remark 3.9. (1) $\mathrm{tr.deg}_{\mathbb{C}}\, \mathbb{C}(f_1, ..., f_n, \exp(f_1), ..., \exp(f_n))$ is the dimension of the Zariski closure of U in $\mathbb{C}^n \times (\mathbb{C}^*)^n$.

(2) $\mathrm{rank}(J)$ is the (analytic) dimension of U.

(3) f_i have no constant terms and they are linearly independent over $\mathbb{Q}$ implies the projection of U to $\mathbb{C}^n$ contains the origin and is not contained in linear subspace defined over $\mathbb{Q}$.

Definition 3.10. For $k, n \geqslant 1$ and $X \subset \mathbb{R}^n$, a *strong k-parametrization of* X is a C^k-map

$$f : (0,1)^m \to \mathbb{R}^n$$

for some $m < n$ with image X, such that $|f^{(\alpha)}(a)| \leqslant 1$ for all $\alpha \in \mathbb{N}^m$ with $|\alpha| \leqslant k$ and all $a \in (0,1)^m$. We also define a *hypersurface* in $\mathbb{R}^n$ of degree $\leqslant e$ to be the zero set in $\mathbb{R}^n$ of a nonzero polynomial with variables $x_1, ..., x_n$ over $\mathbb{R}$ of degree $\leqslant e$.

It will suffice to apply the following two theorems to prove Pila-Wilkie theorem.

Theorem 3.11. Let $n \geqslant 1$ be given. Then for any $e \geqslant 1$ there are $k = k(n,e) \geqslant 1, \epsilon = \epsilon(n,e)$, and $c = c(n,e)$, such that if $X \subset \mathbb{R}^n$ has a strong k-parametrization, then for all T at most cT^ϵ many hypersurfaces in $\mathbb{R}^n$ of degree $\leqslant e$ are enough to cover $X(\mathbb{Q}, T)$, with $\epsilon(n,e) \to 0$ as $e \to \infty$.

Let $E \subset \mathbb{R}^m$ and $X \subset E \times \mathbb{R}^n$. For $s \in E$, we recall the firbre of X at s is

$$X_s := \{a \in \mathbb{R}^n : (s,a) \in X\}.$$

If E and X are definable in the o-minimal expansion $\tilde{\mathbb{R}}$ of $\mathbb{R}$, then the fibres are definable in $\tilde{\mathbb{R}}$.

Proposition 3.12. Let $\pi : \mathbb{R}^{m+n} \to \mathbb{R}^m$ denote the projection. Suppose $\mathfrak{D}$ is a decomposition of $\mathbb{R}^{m+n}$ partitioning X. Then for each $s \in E$,

$$\mathfrak{D}_s := \{C_s : C \in \mathfrak{D}, s \in \pi(C)$$

is a decomposition of $\mathbb{R}^n$ partitioning X_s. This gives in particular a finite bound on the number of definably conncected components of X_s independent of $s \in E$.

Remark 3.13. This assertion is still true if $\mathbb{R}$ is replaced by any o-minimal field.

Theorem 3.14. Let $\tilde{\mathbb{R}}$ be an o-minimal expansion of $\mathbb{R}$ and $E \subset \mathbb{R}^m$ and $X \subset E \times \mathbb{R}^n$ with $n \geqslant 1$ definable in $\tilde{\mathbb{R}}$ such that every fibre X_s is a subset of $[-1,1]^n$ with empty interior. Then for every $k \geqslant 1$ there is an $M \in \mathbb{N}$ such that every fibre X_s is the union of at most M subsets, each having a strong k-parametrization.

From now on we assume $n \geqslant 1$. We begin by stating some elementary facts about X^{alg} and X^{tr} for $X \subset \mathbb{R}^n$. The first one is obvious:

Lemma 3.15. If $X = X_1 \cup \cdots \cup X_m$, then $X^{\text{alg}} \supset X_1^{\text{alg}} \cup \cdots \cup X_m^{\text{alg}}$, and thus

$$X^{\text{tr}} \subset X_1^{\text{tr}} \cup \cdots \cup X_m^{\text{tr}}.$$

Note also that if X is open in $\mathbb{R}^n$, then $X^{\text{tr}} = \varnothing$.

Lemma 3.16. Suppose $S \subset \mathbb{R}^n$ is semialgebraic, $f : S \to \mathbb{R}^m$ is semialgebraic and injective, and f maps the set $X \subset S$ homeomorphically onto $Y = f(X) \subset \mathbb{R}^m$. Then $f(X^{\text{alg}}) = Y^{\text{alg}}$ and thus $f(X^{\text{tr}}) = Y^{\text{tr}}$ (We allow $m = 0$ for latter inclusions).

Proof. It is clear that $f(X^{\text{alg}}) \subset Y^{\text{alg}}$. Also, for any connected infinite semialgebraic set $C \subset Y$, the set $f^{-1}(C) \subset S$ is semialgebraic (since C and f are), contained in X (since f is injective), hence connected and infinite, and thus $f^{-1}(C) \subset X^{\text{alg}}$. This shows $f^{-1}(Y^{\text{alg}}) \subset X^{\text{alg}}$, and thus $f(X^{\text{alg}}) = Y^{\text{alg}}$. $\square$

In order to apply Theorem 3.14, we need to reduce to the case of subsets of $[-1,1]^n$. This is done as follows. For $X \subset \mathbb{R}^n$ and $I \subset \{1,...,n\}$, set

$$X_I := \{a \in X : |a_i| > 1 \text{ for all } i \in I,\ |a_i| \leqslant 1 \text{ for all } i \notin I\}$$

and define the semialgebraic map $f_I : \mathbb{R}^n_I \to \mathbb{R}^n$ by $f_I(a) = b$ where $b_i := a_i^{-1}$ for $i \in I$ and $b_i := a_i$ for $i \notin I$. Thus f_I maps $\mathbb{R}^n_I$ homeomprhically onto its image, a subset of $[-1,1]^n$. If $I = \varnothing$, then f_I is the inclusion map $\mathbb{R}^n_I = [-1,1]^n \to \mathbb{R}^n$. Note that for $a \in \mathbb{Q}^n$ we have $f_I(a) \in \mathbb{Q}^n$ and $H(a) = H(f_I(a))$. Moreover, X is the disjoint union of the sets X_I, and for $Y_I = f_I(X_I)$ we have $Y_I \subset [-1,1]^n$, $Y_I^{\mathrm{tr}} = f_I(X_I^{\mathrm{tr}})$ by Lemma 3.16. Therefore, $N(Y_I^{\mathrm{tr}}, T) = N(X_I^{\mathrm{tr}}, T)$ for all T.

Sketch of the proof: Let $X \subset \mathbb{R}^n$ be definable. We need to show that there are 'few' rational points on X^{tr}. We proceed by induction by n. By Lemma 3.15 we can remove the interior of X in $\mathbb{R}^n$ from X and arrange that X has empty interior. As indicated in the previous discussion we also arrange $X \subset [-1,1]^n$.

Let ϵ be given, and take $e \geqslant 1$ so large that $\epsilon(n,e) \leqslant \epsilon/2$ in Theorem 3.11, and take $k = k(n,e)$. Theorem 3.14 gives $M \in \mathbb{N}$ such that X is a union of at most M subsets, each admitting a strong k-parametrization. Then Theorem 3.11 gives

$$X(\mathbb{Q}, T) \subset \bigcup_{i=1}^{M} \bigcup_{j=1}^{J} H_{ij},$$

where H_{ij} are hypersurfaces in $\mathbb{R}^n$ of degree $\leqslant e$, and $J \in \mathbb{N}$ such that $J \leqslant cT^{\epsilon/2}$ with $c = c(n,e)$ in that theorem. If $a \in X^{\mathrm{tr}}(\mathbb{Q}, T)$ and $a \in H_{ij}$, then clearly $a \in (X \cap H_{ij})^{\mathrm{tr}}$. Thus

$$X^{\mathrm{tr}}(\mathbb{Q}, T) \subset \bigcup_{i=1}^{M} \bigcup_{j=1}^{J} (X \cap H_{ij})^{\mathrm{tr}}(\mathbb{Q}, T).$$

Let H be any hypersurface in $\mathbb{R}^n$ of degree $\leqslant e$. We aim for an upper bound on $N((X \cap H)^{\mathrm{tr}}, T)$ of the form $c_1 T^{\epsilon/2}$ with $c_1 \in \mathbb{R}^{>}$ independent of H and T. This is because if we achieve this, then applying this to the hypersurfaces H_{ij} we obtain

$$N(X^{\mathrm{tr}}, T) \leqslant MJc_1T^{\epsilon/2} \leqslant M \cdot cT^{\epsilon/2} \cdot c_1 T^{\epsilon/2} = Mcc_1 \cdot T^{\epsilon}.$$

Thus we are done. To get this, first we apply the Cell Decomposition Theorem so that we can take semialgebraic cells $C_1, ..., C_L$ in $\mathbb{R}^n$ such that

$$H = C_1 \bigcup \cdots \bigcup C_L.$$

Suppose $C = C_l$ is one of those cells. Then we have a semialgebraic homeomorphism

$$p = p_C : C \to p(C) = p(C_l)$$

onto an open cell $p(C_l)$ in some $\mathbb{R}^{n_l}$ with $n_l < n$. So p maps $X \cap C_l$ homeomorphically onto its image $Y_l \subset p(C_l) \subset \mathbb{R}^{n_l}$. Now p is given by omitting $n - n_l$ of the coordinates, so for $a \in C_l(\mathbb{Q})$ we have $p(a) \in \mathbb{Q}^{n_l}$ and $H(p(a)) \leqslant H(a)$. The hypersurfaces of degree $\leqslant e$ in $\mathbb{R}^n$ belong to a single semialgebraic family, so by Proposition 3.12 we can take $L \leqslant L(e,n)$ with $L(e,n) \in \mathbb{N}^{\geqslant 1}$ depending only on e, n. By Lemma 3.15,

$$(X \cap H)^{\mathrm{tr}} \subset (X \cap C_1)^{\mathrm{tr}} \bigcup \cdots \bigcup (X \cap C_L)^{\mathrm{tr}}.$$

Since the $n_l < n$ we can assume inductively that for all T,

$$N(Y_l^{\mathrm{tr}}, T) \leqslant B_l T^{\epsilon/2},\ l = 1, ..., L$$

with $B_l \in \mathbb{R}^{>}$ independent of T. Hence for all T,

$$N((X \cap C_l)^{\mathrm{tr}}, T) \leqslant B_l T^{\epsilon/2},\ l = 1, ..., L$$

by Lemma 3.16 applied to the maps $p = p_{C_l}$, and thus

$$N((X \cap H)^{\mathrm{tr}}, T) \leqslant (B_1 + \cdots + B_L) T^{\epsilon/2}.$$

Assume we can take $B_1, ..., B_L \leqslant B$ with $B \in \mathbb{R}^{>}$ depending only on X, ϵ, not on $H, Y_1, ..., Y_L$. Then $c_1 := L(e,n)B$ is a positive real number as aimed for.

Remark 3.17. The above sketch is a proof up to the assumption at the end. The assumption can be proved, and we shall not work on the details for the assumption.

We end up our talk with two definable-family versions of Pila-Wilkie theorem. Let $E \subset \mathbb{R}^m$ and $X \subset E \times \mathbb{R}^n$ be definable sets.

Theorem 3.18. Given any ϵ, there is a constant $c = c(X, \epsilon)$ such that for all $s \in E$ and all T we have $N(X_s^{\mathrm{tr}}, T) \leqslant cT^{\epsilon}$.

Theorem 3.19. Given any ϵ, there is a definable set $V = V(X, \epsilon) \subset X$ and a constant $c = c(X, \epsilon)$ such that for all $s \in E$ and all T,

$$V_s \subset X_s^{\mathrm{alg}}, \quad \text{and} \quad N(X_s \backslash V_s, T) \leqslant cT^{\epsilon}.$$

1 Forcing conditions and generic sets

The main idea of forcing is to expand a model of ZF(C) by a single set to force it to have some property. We will start by fixing a base model of ZF(C) $\mathfrak{M}$ (thanks to Godels completeness theorem we know this model exists, when assuming that ZF(C) is consistent). We will be able to expand the model by a special set G, that while it is not in the base model, we can still assume that it exists.
For this we will look at partially ordered sets. Take $\mathfrak{M}$ to be a countable transitive model (inside a bigger model) and look at some partially ordered set $(P, <)$, we will call it a *notion of forcing* and it's element *forcing conditions*. We will call:

- $q \in P$ is *stronger* than $p \in P$ if $q > p$
- $p, q \in P$ are *compatible* if there is some $r \in P$ s.t. $p \geq r \leq q$, otherwise they are *incompatible*.
- $W \subseteq P$ is an *antichain* if all elements of W are pairwise incompatible.
- $D \subseteq P$ is *dense* if for each $p \in P$ there is some $q \in D$ s.t. $q \leq p$

This leads us to the central definition of this talk:

Definition 1. A set $F \subseteq P$ is a *filter* if:

- $F \neq \varnothing$
- if $p \leq q$ and $p \in F$ then $q \in F$
- any $p, q \in F$ are compatible

A filter $G \subseteq P$ is generic over $\mathfrak{M}$ if for any dense subset $D \subseteq P$ s.t. $D \in \mathfrak{M}$ we have $G \cap D \neq \varnothing$

Note that in most cases generic sets over $\mathfrak{M}$ do not exist in $\mathfrak{M}$, however since we have chosen $\mathfrak{M}$ to be a countable transitive model we can always find a generic filter in the larger model:

Lemma 1. *For any partially ordered set $(P, <)$ and any countable collection of dense subsets of P: $\mathcal{D}$, we have some $\mathcal{D}$-generic filter G on P. Furthermore for each $p \in P$ we can choose G s.t. $p \in G$*

Proof. Write $\mathcal{D} = \{D_1, D_2, \dots\}$. Set $p_0 = p$ and take for each $n \in \mathbb{N}_{\geq 1}$: p_n to be s.t. $p_n \leq p_{n-1}$ and $p_n \in D_n$.
Then the set:

$$G = \{q \in P \mid \exists n \in \mathbb{N},\ q \geq p_n\}$$

is a $\mathcal{D}$-generic filter. $\square$

Since we have assumed that the model $\mathfrak{M}$ is countable we can always assume that we have some generic filter outside of our ground model.
We will show that we can always add this generic filter to our model without changing too much:

Theorem 1. *For any transitive model $\mathfrak{M}$ of ZF(C) and any partially ordered set $(P,<)$. If $G \subseteq P$ is a $\mathfrak{M}$-generic filter over P, then there exists a transitive model $\mathfrak{M}[G]$ s.t.*

- *$\mathfrak{M}[G]$ is a model of ZF(C)*
- *$\mathfrak{M} \subseteq \mathfrak{M}[G]$ and $G \in \mathfrak{M}[G]$*
- *$Ord^{\mathfrak{M}} = Ord^{\mathfrak{M}[G]}$*
- *if $\mathfrak{N}$ is a transitive model of ZF s.t. $\mathfrak{M} \subseteq \mathfrak{N}$ and $G \in \mathfrak{N}$ then $\mathfrak{M}[G] \subseteq \mathfrak{N}$*

We will call $\mathfrak{M}[G]$ the generic extension of $\mathfrak{M}$. The model will be constructed s.t. any element of it will have a name in $\mathfrak{M}$. We will construct a forcing language that will contain a name for each element in $\mathfrak{M}[G]$.
However partial orders are not strong enough to formulate this language, so we must turn to complete Boolean algebras instead.

2 Complete Boolean algebras

We will view Boolean algebras as a bounded partially ordered set where each two elements have a supremum and an infimum (a lattice), that fulfill certain distributive properties with each element having a complement. We will say that a Boolean algebra is complete if each set has a supremum and infimum.

Definition 2. A Boolean algebra is a bounded lattice B with maximum 1 and minimum 0 and meet $\vee$ and join $\wedge$, that satisfies:

$$\forall a,b,c \in B,\ a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) \qquad a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$
$$\forall a \in B \exists \neg a \in B,\ a \vee (\neg a) = 1 \qquad a \wedge (\neg a) = 0$$

We will call the algebra complete if each set has a supremum and an infimum.
We will denote $B^+ := B \setminus \{0\}$ (and look at it as a partially ordered set).

We will define the filter and ideal on a Boolean algebra:

Definition 3. An ideal of a Boolean algebra B is a subset $I \subseteq B$ s.t.:

- $0 \in I, 1 \notin I$
- $u, v \in I \implies u \vee v \in I$
- $u \in I, v \leq u \implies v \in I$

a filter is a subset $F \subseteq B$ s.t.

- $1 \in F, 0 \notin F$
- $u, v \in F \implies u \wedge v \in F$
- $u \in F, u \leq v \implies v \in F$

We will call an ideal I a prime ideal if for each $u \in B$ we have $u \in I$ or $\neg u \in I$. Analogously we have say that a filter is an ultrafilter if for each $u \in B$ we have $u \in F$ or $\neg u \in F$.

As Boolean algebras are much more powerful we will do all our constructions on them. However they will still remain general for all partially ordered sets since for any p.o.-set P we have some complete Boolean algebra $B(P)$ that has essentially the same generic filters:

Lemma 2. *For any partially ordered set $(P, <)$ there is some complete Boolean algebra $B(P)$ and a function $e : P \to B(P)^+$ s.t.:*

- $p \leq q \implies e(p) \leq e(q)$
- *p, q are comparable iff $e(p) \wedge e(q) \neq 0$*
- *$\{e(p) \mid p \in P\}$ is dense in $B(P)$*

Furthermore e is definable in $\mathfrak{M}$ and for any set G in P we can define the set $H := \{u \in B(P)^+ \mid \exists p \in G,\ e(p) \leq u\}$ with G being a $\mathfrak{M}$-generic filter iff H is.

So instead of doing the generic extension in P we can do it in $B(P)^+$. Additionally if we say that an ultrafilter G is generic over the base model $\mathfrak{M}$ iff:

$$X \in \mathfrak{M}, X \subseteq G \implies \bigwedge X \in G$$

One can show that G is a generic ultrafilter in B iff it is a generic filter B^+. So instead of working with generic filters over partial orders we can work with generic ultrafilters over complete Boolean algebras.

3 Boolean Valued Models

We will now generalize the notion of the model of ZF(C), by evaluating a formula not in $\{0, 1\}$ but in a Boolean algebra.

Definition 4. For a complete Boolean algebra B, a Boolean-valued model $\mathfrak{U}$ consists of a *Boolean universe* A and two functions from A^2 to B:

$$\|x = y\|, \quad \|x \in y\|$$

that satisfy:

- $\|x = x\| = 1$

- $\|x = y\| = \|y = x\|$
- $\|x = y\| \wedge \|y = z\| \leq \|x = z\|$
- $\|x \in y\| \wedge \|v = x\| \wedge \|w = y\| \leq \|v \in w\|$

we will call this the Boolean value of these formulas.

We can expand the notion of Boolean value to any formula ϕ: $\|\phi\|$ by recursion over the complexity by setting for each $a_1, \dots, a_n \in A$:

- For atomic formulas it has already been defined:
- For negation, conjunction of formulas we set:

$$\|\neg\psi(a_1, \dots, a_n)\| = \neg\|\psi(a_1, \dots, a_n)\|$$
$$\|(\psi \wedge \chi)(a_1, \dots, a_n)\| = \|\psi(a_1, \dots, a_n)\| \wedge \|\chi(a_1, \dots, a_n)\|$$

- For formulas starting with existential quantifier:

$$\|\exists x,\ \psi(x, a_1, \dots, a_n)\| = \bigvee_{a \in A} \|\psi(a, a_1, \dots, a_n)\|$$

We will say that $\phi(a_1, \dots, a_n)$ is *valid* in $\mathfrak{U}$ if $\|\phi(a_1, \dots, a_n)\| = 1$. We can show that the implication $\phi \to \psi$ is valid if $\|\phi\| \leq \|\psi\|$. We can show that the axioms of predicate calculus are all valid. Thus if we have some Boolean-valued in which all axioms of ZF(C) are valid, then for any set theoretical statement σ if we have $\|\sigma\| \neq 0, 1$ then σ is unprovable in ZF(C).
We can transform a Boolean-valued model $\mathfrak{U}$ into a regular model if it has the following property:

Definition 5. A Boolean valued model is *full* if for any formula $\phi(x, x_1, \dots, x_n)$, and any $a_1, \dots, a_n \in A$ there exists an $a \in A$ s.t.

$$\|\phi(a, a_1, \dots, a_n)\| = \|\exists x,\ \phi(x, a_1, \dots, a_n)\|$$

If we have such a model we can for any ultrafilter F of B, define a model $\mathfrak{U}/F$. We will define an equivalence relation $\equiv$ on A by setting:

$$x \equiv y \iff \|x = y\| \in F$$

and a binary relation E on $A/\equiv$ by:

$$[x]E[y] \iff \|x \in y\| \in F$$

Since F is a filter and by the properties of the Boolean-valued model this is well-defined. Furthermore we can show when a statements are true in the model $\mathfrak{U}/F = (A/\equiv, E)$:

Lemma 3. *For any $\mathfrak{U}$, full Boolean-valued model and any filter F, then for any formula $\phi(x_1, \dots, x_n)$ we have:*

$$\mathfrak{U}/F \vDash \phi([a_1], \dots, [a_n]) \iff \|\phi(a_1, \dots, a_n)\| \in F$$

for all $a_1, \dots, a_n \in A$

Proof. For atomic formulas this follows from the definition.
If ϕ is a negation of a conjunction this follows from the following facts:

$$\neg\|\psi\| = \|\neg\psi\| \in F \iff \|\psi\| \notin F$$
$$\|\psi\| \wedge \|\chi\| = \|\psi \wedge \chi\| \in F \iff \|\psi\| \in F \text{ and } \|\chi\| \in F$$

If ϕ is of the form $\exists x \psi(x, \dots)$, we can use that $\mathfrak{U}$ is full and thus we have some $a \in A$ s.t. $\|\psi(a, \dots)\| = \|\exists x \psi(x, \dots)\| = \|\phi\|$ and thus:

$$\|\phi\| \in F \iff (\exists a \in A),\ \|\psi(a, \dots)\| \in F$$

thus by induction the lemma follows. □

Now we have to build a full Boolean-valued model V^B for any complete Boolean algebra B.

4 V^B

To define the Boolean valued model V^B we will effectively look at functions with values in B. Formally we can define for any ordinal α inductively the sets V_α^B as follows:

- $V_0^B = \varnothing$
- $V_{\alpha+1}^B = \{x : y \to B \mid y \subseteq V_\alpha^B\}$
- For any limit ordinal α: $V_\alpha^B = \bigcup_{\beta<\alpha} V_\beta^B$

And so $V^B = \bigcup_{\alpha \in \mathbf{Ord}} V_\alpha^B$. We can assign each element $x \in V^B$ the rank:

$$\rho(x) = \min\{\alpha \mid x \in V_{\alpha+1}^B$$

To define the norm we will first introduce the notation on B:

$$u \implies v := \neg u \vee v$$

an using this we can define:

- $\|x \in y\| = \bigvee_{t \in \mathrm{dom}(y)} (\|x = t\| \wedge y(t))$
- $\|x \subset u\| = \bigwedge_{t \in \mathrm{dom}(x)} (x(t) \implies \|t \in y\|)$
- $\|x = y\| = \|x \subset y\| \wedge \|y \subset x\|$

Verifying that this is indeed a Boolean valued model is tedious so we will only look at the first condition:

Lemma 4.

$$\forall x \in V^B, \ \|x = x\| = 1$$

Proof. We use induction over $\rho(x)$. By definition all we need is to show that $\|x \subset x\| = 1$, i.e. $x(t) \implies \|t \in x\| = 1$ for each $t \in \mathrm{dom}(x)$ which is the same as $x(t) \leq \|t \in x\|$. But since $\rho(t) < \rho(x)$ we have $\|t = t\| = 1$ and thus $x(t) = \|t = t\| \wedge x(t) \leq \|t \in x\|$ □

Similarly we show:

Lemma 5. *V^B is a Boolean-valued model.*

Proof. Exercise. □

Now we need to show that V^B is full and that each of the axioms of ZFC are valid in V^B. We will first show extensionality and then fullness.

Lemma 6. *V^B is extensional*

Proof. We note that showing that V^B is extensional is equivalent to showing that for each $X, Y \in V^B$:

$$\|\forall u(u \in X \leftrightarrow u \in Y)\| \leq \|X = Y\|$$

Note that by our definition of $a \implies b$ we have for $a \leq a'$: $a' \implies b \leq a \implies b$ and thus:

$$\bigwedge_{u \in V^B} (\|u \in X\| \implies \|u \in Y\|) \leq \bigwedge_{u \in V^B} (X(u) \implies \|u \in Y\|)$$

and since left side is equal to $\|\forall u(u \in X \rightarrow u \in Y)\|$ and the right is equal to $\|X \subset Y\|$. By swapping X and Y we then get the desired result. □

For fullness we will need a technical lemma:

Lemma 7. *For a set of pairwise disjoint elements of B: W and any $\{a_u \mid u \in W\} \subseteq V^B$ there exists some $a \in V^B$ s.t.*

$$\forall u \in W u \leq \|a = a_u\|$$

Proof. Define $D = \bigcup_{u \in W} \mathrm{dom}(a_u)$, and for any $t \in D$ $a(t) = \bigvee\{u \wedge a_u(t) \mid u \in W\}$. Since the u's are disjoint we have $u \wedge a(t) = u \wedge a_u(t)$. Thus $u \leq (a(t) \implies a_u(t))$ and $u \leq (a_u(t) \implies a(t))$ so $u \leq \|a = a_u\|$.

□

This allows us to show that:

Lemma 8. *V^B is full.*

Proof. We note that being full means that for each formula $\phi(x,\dots)$ there is some $a \in V^B$ s.t.

$$\|\phi(a,\dots)\| = \|\exists x\phi(x,\dots)\|$$

By definition $\leq$ holds for each a. To find some $a \in V^B$ s.t. $\geq$ holds, we define $u_0 := \|\exists x\phi(x,\dots)\|$ and

$$D = \{u \in B \mid \exists a_u \in V^B,\ u \leq \|\phi(a_u,\dots)\|\}$$

Note that D is dense and open below u_0, take W to be a maximal set of pairwise disjoint elements of D (using AC!). We have $\bigvee\{u \mid u \in W\} \geq u_0$. By the above lemma we have some $a \in V^B$ s.t. $u \leq \|a = a_u\|$ so for each $u \in W$ we have:

$$u \leq \|a = a_u\| \wedge \|\phi(a_u,\dots)\| = \|\phi(a,\dots)\|$$

showing that $u_0 \leq \|\phi(a,\dots)\|$. □

We note that we can embed every set in $\mathfrak{M}$ in V^B by defining:

Definition 6.

- $\hat{\varnothing} = \varnothing$
- $\hat{x} : \begin{cases} \{\hat{y} \mid y \in x\} \to B \\ \hat{y} \mapsto 1 \end{cases}$

and for a generic ultrafilter G we can define the canonical name for it as

$$\dot{G} : \begin{cases} \{\hat{u} \mid u \in B\} \to B \\ \hat{u} \mapsto u \end{cases}$$

The proof that in V^B each axiom of ZFC is valid would be too long for this talk so we will omit it:

Theorem 2. *In V^B the axioms of ZFC are valid.*

So when we take M^B to be the Boolean valued model with universe V^B we get a Boolean valued model where ZFC is valid. With a generic ultrafilter we can do some interesting stuff here.

5 The Forcing Theorem and the Generic Model Theorem

Take some $\mathfrak{M}$-generic ultrafilter G in in B then we can interpret each $x \in M^B$ as follows:

Definition 7. For each $x \in M^B$ define x^G recursively over $\rho(x)$:

- $\varnothing^G = \varnothing$
- $x^G = \{y^G \mid x(y) \in G\}$

We get:

$$M[G] = \{x^G \mid x \in M^B\}$$

This gives us a model of which we can show that:

Lemma 9. *For all names $x, y \in M^B$:*

- $x^G \in y^G \iff \|x \in y\| \in G$
- $x^G = y^G \iff \|x = y\| \in G$

Proof. We will prove those two simultaneously by induction over $(\rho(x), \rho(y))$

$$\begin{aligned}\|x \in y\| \in G &\iff \exists t \in \mathrm{dom}(y)(y(t) \in G \text{ and } \|x = t\| \in G)\\ &\iff \exists t \in \mathrm{dom}(t)(y(t) \in G \text{ and } x^G = t^G)\\ &\iff x^G \in \{t^G \mid y(t) \in G\}\\ &\iff x^G \in y^G\end{aligned}$$

$$\begin{aligned}\|x \subset y\| \in G &\iff \bigwedge_{t \in \mathrm{dom}(x)} (x(t) \implies \|t \in y\|) \in G\\ &\iff \forall t \in \mathrm{dom}(x)(x(t) \in G \implies \|t \in y\| \in G)\\ &\iff \forall t,\ (x(t) \in G \implies t^G \in y^G)\\ &\iff \{t^G \mid x(t) \in G\} \subset y^G\\ &\iff x^G \subset y^G\end{aligned}$$

□

This shows that there is an equivalence between $M[G]$ and M^B/G and thus we get:

Theorem 3 (Forcing Theorem)**.** *For any M-generic ultrafilter on Bwe have for any $x_1, \ldots, x_n \in M^B$:*

$$MG \vDash \phi(x_1^G, \ldots x_n^G) \iff \|\phi(x_1, \ldots, x_n)\| \in G$$

Proof. Since $M[G]$ is equivalent to M^B/G this follows from a previous result. □

a consequence of this is

Corollary 1. *$M[G]$ is a model of ZFC*

Now to show the generic model theorem we have to show that:

Lemma 10.

- *$\mathfrak{M} \subset M[G]$ and the models have the same ordinals*
- *$G \in M[G]$ and for any transitive ZFC model N s.t. $M \subset N$ and $G \in N$ we have $M[G] \subseteq N$*

Proof.

- For each $x \in M$ we have $\hat{x}^G = x$, hence $M \subset M[G]$. To show that every ordinal in $M[G]$ is in M one can note that being an ordinal is a relatively simple condition.
- Since $\dot{G}^G = G$ we have $G \in M[G]$. Additionally if $N \supset M$ and $G \in N$ we can construct $M[G]$ in N and thus $M[G] \subseteq N$

$\square$

Model Theory Week 13 - Forcing and CH

June 1, 2023

1 From last week

Let $\mathbf{M}$ be our countable, ground model of ZF and $\mathbf{N}$ a bigger model of ZFC.

Let $(P, <)$ be a partially ordered set. We will call it a *notion of forcing* and it's element *forcing conditions*. We will call:

- $q \in P$ is *stronger* than $p \in P$ if $q > p$
- $p, q \in P$ are *compatible* if there is some $r \in P$ s.t. $p \geq r \leq q$, otherwise they are *incompatible*.
- $W \subseteq P$ is an *antichain* if all elements of W are pairwise incompatible.
- $D \subseteq P$ is *dense* if for each $p \in P$ there is some $q \in D$ s.t. $q \leq p$

Definition 1.1. A set $F \subseteq P$ is a *filter* if:

- $F \neq \emptyset$
- if $p \leq q$ and $p \in F$ then $q \in F$
- any $p, q \in F$ are compatible

A filter $G \subseteq P$ is generic over $\mathbf{M}$ if for any dense subset $D \subseteq P$ s.t. $D \in \mathbf{M}$ we have $G \cap D \neq \emptyset$

Lemma 1.2. *For any partially ordered set $(P, <)$ and any countable collection of dense subsets of P: $\mathcal{D}$, we have some $\mathcal{D}$-generic filter G on P. Furthermore for each $p \in P$ we can choose G s.t. $p \in G$.*

Theorem 1.3. *For any transitive model $\boldsymbol{M}$ of ZF(C) and any partially ordered set $(P, <)$. If $G \subseteq P$ is a $\boldsymbol{M}$-generic filter over P, then there exists a transitive model $\boldsymbol{M}[G]$ s.t.*

- *$\boldsymbol{M}[G]$ is a model of ZF(C)*
- *$\boldsymbol{M} \subseteq \boldsymbol{M}[G]$ and $G \in \boldsymbol{M}[G]$*
- *$Ord^{\boldsymbol{M}} = Ord^{\boldsymbol{M}[G]}$*
- *if $\boldsymbol{N}$ is a transitive model of ZF s.t. $\boldsymbol{M} \subseteq \boldsymbol{N}$ and $G \in \boldsymbol{N}$ then $\boldsymbol{M}[G] \subseteq \boldsymbol{N}$*

2 ZFC+CH

We want to add a function $f : \mathcal{P}(\mathbb{N}) \to \omega_1$ a bijection to our ground model $\mathbf{M}$, making the CH true for the new model.

FIRST CONSTRUCTION (not the right one)

Let P_1 be the partially ordered set of all finite, partial bijections between $\mathcal{P}(\mathbb{N})$ and ω_1, ordered by $\subset$. Let G be a generic filter of P_1. We then obtain $f = \cup G$ a function from $\mathcal{P}(\mathbb{N})$ to ω_1. Notice that:

- By construction, it is an injective map from a subset of $\mathcal{P}(\mathbb{N})$ to a subset of ω_1.
- For every $A \in \mathcal{P}(\mathbb{N})$, as the set $D_A = \{f \in P_1;\ A \text{ is in the domain of } f\}$ is dense in P_1, it follows that A is in the domain of f. Hence the domain of f is $\mathcal{P}(\mathbb{N})$.

- In a similar way, for every $\alpha \in \omega_1$, as the set $D_\alpha = \{f \in P_1;\ \alpha \text{ is in the range of } f\}$, it follows that α is in the range of f. Hence f is surjective.

It is already known that $\mathbf{M}$ and $\mathbf{M}[G]$ have the same ordinals, but not always the case that the cardinals stay the same. More bijections on ordinals can show up and 'collapse' some cardinals. We now need to show that the ω_1 from $\mathbf{M}$ is still the first uncountable ordinal in $\mathbf{M}[G]$.

On the other hand, the property of being a set of natural numbers is universal. So every set of natural numbers in $\mathbf{M}$ continues to be a set of natural numbers in $\mathbf{M}[G]$. That is, $\mathcal{P}(\mathbb{N})^{\mathbf{M}} \subset \mathcal{P}(\mathbb{N})^{\mathbf{M}[G]}$. (Still, new set of natural numbers might show up. In the construction above we can build a set of natural numbers that is not in $\mathbf{M}$, which is the main problem).
SECOND CONSTRUCTION

Let P_2 be the set of all $f \in \mathbf{M}$ such that f is a bijection between a countable subset of $\mathcal{P}(\mathbb{N})$ and a countable subset of ω_1. The key property on this case is the following:

Definition 2.1. A set P is ω-closed if the union of every infinite countable increasing sequence $p_0 \subseteq p_1 \subseteq p_2 \subseteq \ldots$ of elements of P belongs to P.

It is easy to see that P_2 satisfy such property in relation to $\mathbf{M}$. This property ensures that no new function with domain $\mathbb{N}$ appear in $\mathbf{M}[G]$.

Lemma 2.2. *Let P be a forcing notion such that P is ω-closed in $\mathbf{M}$, let G be a generic filter of P, and let $X \in \mathbf{M}$. Then any function $f : \mathbb{N} \to X$ in $\mathbf{M}[G]$ is already in $\mathbf{M}$.*

We can then prove:

Theorem 2.3. *Let P be the set of all $f \in \mathbf{M}$ such that $M \models$ 'f is a bijection between a countable subset of $\mathcal{P}(\mathbb{N})$ and a countable subset of ω_1' and let G be a generic filter of P. Then the continuum hypothesis holds in $\mathbf{M}[G]$.*

Proof. According to the lemma, $\mathbf{M}[G]$ does not introduce any new functions with domain $\mathbb{N}$. Since the subsets of $\mathbb{N}$ correspond to functions from $\mathbb{N}$ into $\{0, 1\}$, it follows that $\mathbf{M}[G]$ does not introduce any new subsets of $\mathbb{N}$, that is, $\mathcal{P}(\mathbb{N})^{\mathbf{M}} = \mathcal{P}(\mathbb{N})^{\mathbf{M}[G]}$. Also, $\mathbf{M}[G]$ cannot introduce a function from $\mathbb{N}$ onto $\omega_1^{\mathbf{M}}$, which means that $\omega_1^{\mathbf{M}}$ must remain uncountable in $\mathbf{M}[G]$. That is, $\omega_1^{\mathbf{M}[G]} = \omega_1^{\mathbf{M}}$.

By (Theorem 1, last week) we have that $G \in \mathbf{M}[G]$. Thus is the function $f = \cup G$, which is a bijection between a subset of $\mathcal{P}(\mathbb{N})^{\mathbf{M}}$ and a subset of $\omega_1^{\mathbf{M}}$. By the same argument as in the first construction, such f is a bijection in $\mathbf{M}[G]$, proving that the continuum hypothesis is true in such model. □

3 ZFC+¬CH

To prove that there exists a model with ¬CH we have to work in a different way to what we had on the other case. For this one we want to build a bijection from $\mathcal{P}(\mathbb{N})$ to ω_2 in such a way that $\mathbf{M}[G]$ has the same cardinals as $\mathbf{M}$.

Definition 3.1. A join-antichain in a set P is a family of pairwise imcompatible elements. P is c.c.c. (countable chain condition) if every join-antichain in P is countable.

This condition allow us to prove the following:

Lemma 3.2. *Let P be a forcing notion such that $\mathbf{M} \models$ 'P is c.c.c.', let G be a generic filter of P, let $X, Y \in \mathbf{M}$, and let $f : X \to Y$ be a function in $\mathbf{M}[G]$. Then there is a function $g : X \to \mathcal{P}(Y)$ in $\mathbf{M}$ such that for each $x \in X$ we have $f(x) \in g(x)$ and $\mathbf{M} \models$ '$g(x)$ is countable'.*

Theorem 3.3. *Let P be a forcing notion such that $\mathbf{M} \models$ 'P is c.c.c.' Then P preserve cardinals.*

We still have to show that the forcing notion we plan to use is c.c.c. according to $\mathbf{M}$. This will be done with the following lemma, which is a theorem of set theory.

Lemma 3.4 (Δ-system lemma)**.** *Let X be an uncountable family of finite sets. Then there is an uncountable subset $Y \subset X$ and a set R such that $A \cap B = R$ for any distinct $A, B \in Y$.*

We now build the forcing notion in $\mathbf{M}$ and the desired generic filter G. Instead of directly introducing a bijection between $\mathcal{P}(\mathbb{N})$ and, say, ω_2, we will aim to introduce a family of ω_2 distinct subsets of $\mathbb{N}$. Such a family can be encoded as all possible functions from $\mathbb{N} \times \omega_2$ into $\{0, 1\}$.

Our set P of functions is the set of all finite, partial functions from $\mathbb{N} \times \omega_2$ into $\{0, 1\}$.

Theorem 3.5. *Let P be as above and G any generic filter of P. Then the continuum hypothesis fails in $\boldsymbol{M}[G]$.*

Proof. We first need to show P is c.c.c. in $\mathbf{M}$. Let S be an uncountable subset of P and let $X = \{\mathrm{dom}(f);\ f \in S\}$. Observe that X must be uncountable, because any finite set supports only finitely many functions into $\{0, 1\}$, and thus any countable family of finite sets can be the domains of only countably many functions into $\{0, 1\}$. By the Δ-system lemma, there exists $Y \subset X$ and a finite set $R \subset \mathbb{N} \times \omega_2$ such that $A \cap B = R$ for any distinct $A, B \in Y$. Form a corresponding infinite subset $T \subseteq S$ by selecting, for each $A \in Y$, one function in S whose domain is A. We then have $\mathrm{dom}(f) \cap \mathrm{dom}(g) = R$ for any distinct $f, g \in T$. But there are only finitely many functions from R into $\{0, 1\}$, so there must exist some $f, g \in T$ such that $f|_R = g|_R$. These two functions are therefore compatible, so we have shown that any uncountable subset of P constains compatible elements.

It follows from Theorem 3.3 that $\omega_2^{\mathbf{M}[G]} = \omega_2^{\mathbf{M}}$. By (Theorem 1, last week) we have that $G \in \mathcal{M}[G]$. Thus $f = \cup G$ is a function from a subset of $\mathbb{N} \times \omega_2^{\mathbf{M}}$ into $\{0, 1\}$, and it belongs to $\mathbf{M}[G]$. Moreover, for each $\langle n, \alpha \rangle$ belongs to $\mathbf{M}$ and is dense, so the domain of f is all of $\mathbb{N} \times \omega_2^{\mathbf{M}}$.

For each ordinal $\alpha < \omega_2$ define

$$A_\alpha = \{n \in \mathbb{N};\ f(n, \alpha) = 1\}.$$

For any distinct $\alpha, \beta < \omega_2$, the set $D_{\alpha,\beta}$ of all $f \in P$ for which there exists $n \in \mathbb{N}$ such that $\langle n, \alpha \rangle$, $\langle n, \beta \rangle \in \mathrm{dom}(f)$ and $f(n, \alpha) \neq f(n, \beta)$ is dense and belongs to $\mathbf{M}$; since G is generic, it must intersect each $D_{\alpha,\beta}$, and this shows that the sets A_α are distinct. Since these sets can be constructed from f in $\mathbf{M}[G]$, it follows that $\mathbf{M}[G] \models$'there are $\aleph_2^{\mathbf{M}[G]}$ distinct subsets of $\mathbb{N}$'. Thus the continuum hypothesis fails in $\mathbf{M}[G]$. □

4 Independence from ZF

The following can be proven to be independent from ZF or ZFC using a similar method:

AC Axiom of choice.

GCH Generalized continuum hypothesis:
$|\mathcal{P}(\omega_\alpha)| = |\omega_{\alpha+1}|$

SP Suslin problem:
Given a non-empty totally ordered set R with the four properties:

- R does not have a least nor a greatest element;
- the order on R is dense;
- the order on R is complete, in the sense that every non-empty bounded subset has a supremum and an infimum; and
- every collection of mutually disjoint non-empty open intervals in R is countable (which is the c.c.c. condition for the order topology of R),

is R necessarily order-isomorphic to the real line $\mathbb{R}$? (Such R is known as Suslin line and the Suslin problem ask if it exists or not).

MA Martin's Axiom:
We say that a cardinal κ has property MA if for any partial order P satisfying the c.c.c. and any family D of dense subset of P such that $|D| < \kappa$, there is a filter F on P such that $F \cap d$ is non-empty for every d in D.
The Martin's axiom states that for every $\kappa < \mathfrak{c}$, κ has property MA, where $\mathfrak{c}$ is the cardinality of the continuum.

$\diamond$ The Diamond principle.

ACons Axiom of constructibility:
Every set is constructible, that is, every set can be built from the empty set via some steps.

The following order shows which of the axioms above implies some of the others.

$$\begin{tikzcd}
 & \mathrm{AC} & & \\
\mathrm{ACons} \arrow[ur] \arrow[r] \arrow[dr] & \mathrm{GCH} \arrow[r] & \mathrm{CH} \arrow[r] & \mathrm{MA} \\
 & \diamond \arrow[r] & \mathrm{SP} &
\end{tikzcd}$$

$$\mathrm{MA} + \neg\mathrm{CH} \longrightarrow \neg\mathrm{SP}$$